

Highlights

Optimization of a Recycling N-Stage Activated Sludge System with a Clarifier using A Physically Constrained Surrogate

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- One surrogate represents the complete recycling N -stage activated sludge system.
- A unique scaled- L_2 QP enforces mixer, reactor, and Clarifier physics.
- Ten Clarifier-layer solids states make the SRT inventory physically traceable.
- A full-rank scaled quadratic basis gives one fixed multiresponse fit.
- Mechanistic simulations separate prediction fidelity from decision quality.

Optimization of a Recycling N-Stage Activated Sludge System with a Clarifier using A Physically Constrained Surrogate

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ABSTRACT

Activated sludge design and operation couple biological conversion, hydraulic staging, mixed-liquor recycle, return activated sludge, sludge wasting, and secondary clarification. Repeated solution of this closed recycle network is too costly for a large bounded search. This study develops a bilevel formulation for a prescribed N -stage activated sludge system with a Clarifier; N is a topology parameter, and the case study fixes $N = 5$. A dimensionless second-order surrogate predicts the inlet mixer, every reactor outlet, the Clarifier outlet component mass flows, and the ten-layer solids profile. A strictly convex lower-level quadratic program then returns one unique corrected network state satisfying recycle mixing, reactor stoichiometric invariants, Clarifier component balances, soluble pass-through, particulate clarification and densification directions, layer-inventory consistency, and non-negativity. The upper level jointly selects reactor volume through nominal hydraulic retention time and selects aeration, mixed-liquor recycle, return-sludge flow, and waste-sludge flow. Training uses a fixed, full-rank standardized multiresponse least-squares estimator on a unique quadratic basis. A 4,000-state set held out from the development fit must pass predictive, numerical, and correction-reliance criteria before the 20,000-state production refit is frozen. The complete mechanistic, training, deployment, numerical-acceptance, and validation definitions are contained in this article. Mechanistic comparison uses a coupled ASM2d-TSN–multilayer-Clarifier model and distinguishes prediction fidelity from the finite-budget mechanistic incumbent gap. The result is a process-level surrogate whose stated physical guarantees and limitations can be traced stream by stream through the activated sludge system.

Nomenclature

Only nonstandard mathematical variables, sets, and operators are listed. Component names and process abbreviations are defined when introduced. All inequalities between vectors are componentwise. Whenever a numerical scaling expression compares a dimensional coordinate with a bare 1, the coordinate is first expressed in its tabulated unit; the 1 therefore means one of that stated unit. This convention creates dimensionless residuals without adding quantities of unlike dimensions.

$A \in \mathbb{R}^{K \times F}$	Full-row-rank stoichiometric-invariant operator
A_{cl}	Clarifier surface area
a_i	Dimensionless aeration setting in CSTR i
$\hat{C}$	Frozen standardized second-order coefficient matrix
$c_i \in \mathbb{R}_+^F$	Component concentration leaving CSTR i
$c_E, c_U \in \mathbb{R}_+^F$	Clarifier overflow and underflow concentrations
D_b, D_χ, D_g, D_T	Positive scales for balances, states, inequalities, and targets
D_θ	Diagonal upper-level decision-range matrix
d_i	Recycle-adjusted dilution rate in CSTR i , d^{-1}
d_ϕ, L, R, R_g	Feature, state, equality-row, and inequality-row dimensions
$\epsilon_{mach} = 2^{-52}$	IEEE 754 binary64 machine epsilon
E_S, E_X	Selectors for soluble and particulate ASM components
F, K, n_θ, N, P	Numbers of components, invariants, decisions, CSTRs, and processes
$g_E = q_E c_E, g_U = q_U c_U$	Fresh-flow-normalized Clarifier outlet component mass flows

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H	Total nominal hydraulic retention time based on fresh flow, h
$\mathcal{H}(\vartheta), \mathcal{G}(\vartheta)$	Closed-loop equality and inequality operators
I_{comp}	Map from components to COD, TN, TP, and TSS
$J(\vartheta, \chi)$	Scalar upper-level engineering objective
$m \in \mathbb{R}_+^F$	Mixed component state entering CSTR 1
$\overline{M}_{\text{cl}} = M_{\text{cl}}/Q_0$	Fresh-flow-normalized Clarifier solids inventory
N_ℓ	Number of Clarifier layers
α_Q	Non-negative normalized effluent-quality weights
q_P, q_C, q_E, q_U	Flow ratios through the train, Clarifier, overflow, and underflow
Q_0	Fresh influent volumetric flow
r_I, r_R, w	MLR, RAS, and WAS flow ratios relative to Q_0
S_ϑ, S_x, S_ψ	Positive design, influent, and feature scaling matrices
s_ℓ	TSS concentration in Clarifier layer ℓ
$t_X \in \mathbb{R}^F$	Column vector whose transpose maps components to TSS
T	Linear map selecting upper-level effluent targets
$\tau_i = 24V_i/Q_0$	Fresh-flow HRT assigned to CSTR i , h
ϑ	Upper-level design-and-operation vector
$x \in \mathbb{R}_+^F$	Known fresh-influent component vector
z	Normalized upper-search coordinate
χ	Stacked mixer, reactor, outlet-flow, and Clarifier-layer state
η_j, δ_j	Underflow recovery and densification factors for component j
$v \in \mathbb{R}^{P \times F}$	ASM2d-TSN stoichiometric matrix
$\phi(\vartheta, x)$	Dimensionless unique-monomial feature vector
ϱ_i	Mechanistic reaction-rate vector in CSTR i
ω_i	Net oxygen-transfer rate in CSTR i
$\mu_\vartheta, \mu_x, \mu_\psi, \mu_\chi$	Coordinate centers calculated from the applicable fitting rows
$\mathcal{R}_s$	Mechanistic decision regret in validation scenario s
$\otimes$	Kronecker product

1. Introduction

Activated sludge plants are recycling systems. Mixed liquor may be returned to an upstream anoxic zone to make oxidized nitrogen available for denitrification, while settled biomass is returned from the Clarifier to maintain the biological inventory. A waste stream controls solids age, and the Clarifier overflow is the treated discharge. These streams are familiar individually, but changing one changes the hydraulic residence time, solids loading, and composition seen elsewhere in the loop.

Detailed activated sludge models represent these interactions through nonlinear reaction rates and clarification fluxes. They are indispensable for process analysis, yet repeated closed-loop steady-state solution can dominate the cost of optimization (de Araujo et al., 2013). Surrogate models can reduce that cost (Bhosekar and Ierapetritou, 2018), and component-space constraints can preserve mass conservation and non-negativity in activated-sludge regression (Selerio Jr., 2026). A complete recycling plant creates a broader risk: the optimizer can seek a small negative concentration, a broken recycle balance, or a favorable solids inventory that would never occur in the plant.

The recycle topology adds a second difficulty. If the reactors and Clarifier are predicted independently, their responses must be iterated around the MLR and RAS loops. A unique answer from each unit does not imply a unique closed-loop fixed point. The complete plant state is therefore represented by one system-level response and reconciled by one explicitly physical calculation. The response uses a unique-monomial quadratic basis and a single coefficient matrix whose identifiability is checked directly from the standardized design.

The resulting contribution is a process-level bilevel model (Colson et al., 2007). The lower level is a strictly convex projection that has one answer for every admissible flow vector. The upper level compares feasible design-and-operation candidates using clarified effluent quality, reactor volume, aeration, pumping, and wasted-solids terms. The Clarifier solids inventory is calculated from predicted layer concentrations. The article defines the mechanistic generator, estimator, projection, trust screens, search, and validation route as one self-contained calculation.

The symbol N in the title denotes a prescribed number of biological stages; it is not an integer decision in the present optimization. The derivation applies to any fixed N , whereas the case study fixes five equal-volume CSTRs, two unaerated followed by three aerated. In the following sections, the equations are introduced in the order that an engineer can follow the water: fresh feed and recycles meet, mixed liquor passes through the reactors, solids and liquid divide in the Clarifier, and only then is the response scored.

2. Theory and Calculation

2.1. Walking Through the Recycling Process

Figure 1 is the physical picture behind every subsequent matrix. Fresh wastewater enters the mixer from the left. Both recycles return to that same point. The combined stream passes once through the N CSTRs. MLR is removed from the last-stage outlet before clarification; the remaining mixed liquor enters the Clarifier. Clarifier underflow is split into RAS and WAS without another composition-changing step, so RAS and WAS share the state c_U .

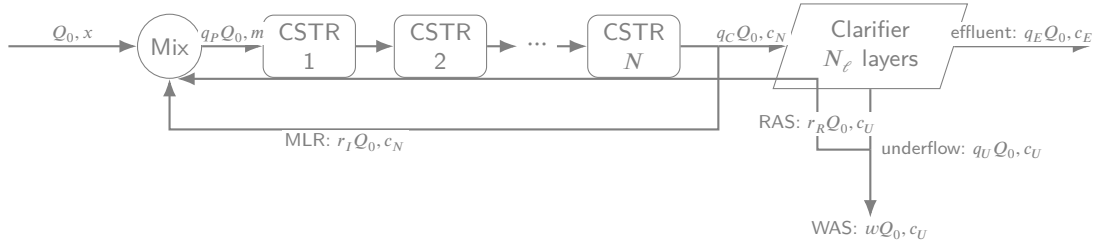


Figure 1: Stream-by-stream picture of the prescribed recycling N -stage activated sludge system. The MLR branch is taken before the Clarifier; RAS and WAS are hydraulic splits of one Clarifier underflow.

The fresh flow Q_0 is a convenient common ruler for the three manipulated side streams. Define

$$r_I = \frac{Q_I}{Q_0}, \quad r_R = \frac{Q_R}{Q_0}, \quad w = \frac{Q_W}{Q_0}, \quad (1)$$

where r_I is the MLR ratio, r_R is the RAS ratio, and w is the WAS ratio. The domain requires $r_I \geq 0$, $r_R > 0$, and $0 \leq w < 1$. These restrictions are not cosmetic: they keep every outlet flow and every concentration recovery division positive.

Starting at the mixer and following the arrows in Figure 1 gives four dimensionless throughflows:

$$\underbrace{q_P}_{\text{through CSTRs}} = 1 + r_I + r_R, \quad \underbrace{q_C}_{\text{into Clarifier}} = 1 + r_R, \quad \underbrace{q_U}_{\text{underflow}} = r_R + w, \quad \underbrace{q_E}_{\text{effluent}} = 1 - w. \quad (2)$$

The first identity counts fresh feed plus both returns. Removing MLR from the last-stage outlet leaves $q_C Q_0$. The Clarifier then divides that stream into $q_E Q_0$ and $q_U Q_0$, and the definitions show directly that $q_C = q_E + q_U$. This small hydraulic calculation is used repeatedly below; it is what makes RAS an internal stream and WAS an external loss.

Let m denote the mixer outlet, c_i the outlet of CSTR i , and c_U the Clarifier underflow. The mass entering the first tank must equal the fresh-feed mass plus the two returned masses, component by component:

$$q_P m = x + r_I c_N + r_R c_U. \quad (3)$$

Every term has the form “flow relative to Q_0 times concentration.” The factor Q_0 therefore cancels. Equation (3) is an identity, not a relationship to be learned. A topology with MLR withdrawn from another stage uses that stage state in place of c_N .

For a general prescribed topology, collect the stage volumes, aeration settings, and recycle decisions as

$$\vartheta = (\tau_1, \dots, \tau_N, a_1, \dots, a_N, r_I, r_R, w)^\top \in \mathcal{V}, \quad \tau_i = \frac{24V_i}{Q_0}. \quad (4)$$

Here τ_i is the nominal HRT in hours that tank i would provide at fresh flow alone. At fixed Q_0 , choosing τ_i chooses volume, $V_i = Q_0\tau_i/24$; it is therefore a design decision, not merely an operating knob. The liquid actually traverses the tank at $q_P Q_0$, so its one-pass residence time and dilution rate are

$$h_i = \frac{\tau_i}{q_P}, \quad d_i = \frac{24q_P}{\tau_i} \text{ d}^{-1}. \quad (5)$$

An engineer can read Equation (5) directly: increasing either recycle while tank volume is fixed moves more water through the same vessel and shortens one pass. The surrogate must see this coupling because the biology does.

2.2. From CSTR Balances to Reaction-Free Inventory Rails

The next task is to carry physically meaningful information through all N biological stages without asking the fast surrogate to reproduce every nonlinear rate. For CSTR i , steady accumulation is zero, so the difference between outlet and inlet transport must be supplied by reactions and oxygen transfer:

$$d_i(c_i - c_{i-1}) = v^\top \rho_i + \omega_i e_{S_O}, \quad c_0 = m. \quad (6)$$

The F entries of $c_i - c_{i-1}$ are component changes across the tank. The P entries of ρ_i are the ASM2d-TSN process rates, and the columns of $v^\top$ translate those rates into component production or consumption. The last term adds oxygen through aeration; e_{S_O} selects only the dissolved-oxygen coordinate.

The rate vector is precisely the expensive, nonlinear object that should not be carried into every surrogate evaluation. What is needed instead is a set of combinations of components that no modeled reaction or oxygen transfer can change. Appendix A.2 constructs five physical conservation rows in B and normalizes each row separately to form $A \in \mathbb{R}^{K \times F}$. Separate normalization keeps the inert-soluble, nitrogen, phosphorus, alkalinity, and metal inventories as recognizable rails. Its required checks are

$$Av^\top = 0, \quad Ae_{S_O} = 0, \quad \text{rank}(A) = K, \quad \|A_{k,:}\|_2 = 1. \quad (7)$$

The first equality says that each row of A annihilates every reaction source; the second prevents external oxygen transfer from masquerading as a conserved inventory; full row rank removes duplication, and the final relation fixes each rail's scale without mixing it with another inventory.

Premultiplying Equation (6) by A removes both source terms. Because $d_i > 0$, the remaining statement is simply

$$Ac_1 = Am, \quad Ac_i = Ac_{i-1}, \quad i = 2, \dots, N. \quad (8)$$

These equalities are useful to picture as K rails extending through the train: individual COD, nitrogen, phosphorus, and biomass components may move between forms, but the selected stoichiometric inventories cannot jump from one tank to the next. They need not equal Ax , because RAS changes the material circulating inside the plant. The fresh-feed relation appears only after the external Clarifier and WAS boundaries are closed.

2.3. Following Components Through the Clarifier

The Clarifier changes where material leaves, not how much of a non-reactive component exists. Its component balance is therefore

$$q_C c_N = q_E c_E + q_U c_U. \quad (9)$$

The small underflow ratio can make c_U numerically large even when its component mass is ordinary. It is more stable and more intuitive to predict the outlet masses normalized by fresh flow:

$$g_E = q_E c_E, \quad g_U = q_U c_U. \quad (10)$$

Equation (9) then becomes

$$g_E + g_U = q_C c_N. \quad (11)$$

Once q_E and q_U are known, the outlet concentrations are recovered uniquely as $c_E = g_E/q_E$ and $c_U = g_U/q_U$. Thus, non-negative mass-flow coordinates also give non-negative outlet concentrations.

Clarification must not be represented as solids generation. The TSS map satisfies $t_X \in \mathbb{R}_+^F$, $E_S t_X = 0$, and $\|t_X\|_1 > 0$. For a particulate component j whose Clarifier-feed mass exceeds the denominator tolerance defined in the numerical contract, define the underflow recovery and concentration ratio by

$$\eta_j = \frac{g_{U,j}}{q_C c_{N,j}}, \quad \delta_j = \frac{c_{U,j}}{c_{N,j}} = \eta_j \frac{q_C}{q_U}. \quad (12)$$

The first factor, η_j , is the fraction of incoming particulate mass sent downward. Non-negativity and Clarifier balance give $0 \leq \eta_j \leq 1$; the deployed densification rule sharpens this to $q_U/q_C \leq \eta_j \leq 1$ and therefore $1 \leq \delta_j \leq q_C/q_U$. The factor q_C/q_U expresses how recovered mass is concentrated into a smaller water stream. When the feed denominator is below tolerance, neither ratio is identifiable and it is reported as unavailable.

The soluble and particulate coordinates require different physical rules. Let E_S select the ten soluble components and E_X the ten particulate components. Ideal soluble pass-through means that the underflow carries the same soluble concentration as the feed:

$$E_S g_U = q_U E_S c_N. \quad (13)$$

Together with Equation (11), this also gives $E_S c_E = E_S c_U = E_S c_N$. For particulates, the mechanistic Clarifier moves a common floc solids flux. The convex correction enforces the physically necessary direction of that movement component by component:

$$E_X g_U \geq q_U E_X c_N. \quad (14)$$

Equivalently, every particulate underflow concentration is at least its feed concentration. Because $g_E \geq 0$ and Equation (11) holds, recovery cannot exceed the incoming particulate mass. The mechanistic common-recovery relation is evaluated as a diagnostic; it is not falsely claimed as an exact linear-QP guarantee.

A single learned Clarifier inventory would still leave the optimizer a loophole: that scalar could change SRT without being tied to any layer. The state is therefore made more physical by predicting the N_ℓ layer TSS concentrations

$$s = (s_1, \dots, s_{N_\ell})^\top, \quad (15)$$

ordered from the top water surface to the bottom hopper. The two end layers are tied to the outlet TSS values,

$$q_E s_1 = t_X^\top g_E, \quad q_U s_{N_\ell} = t_X^\top g_U, \quad (16)$$

and each interior layer is kept within the physically interpretable top-to-bottom envelope,

$$s_1 \leq s_\ell \leq s_{N_\ell}, \quad \ell = 2, \dots, N_\ell - 1. \quad (17)$$

The Clarifier solids inventory is now a consequence of a pictured profile rather than an independent output:

$$M_{cl} = \sum_{\ell=1}^{N_\ell} V_{cl,\ell} s_\ell, \quad \overline{M}_{cl} = \frac{M_{cl}}{Q_0}. \quad (18)$$

The stream bookkeeping can now be closed around the entire plant. Premultiply the mixer balance by A , use the reactor rails in Equation (8), substitute the Clarifier balance, and cancel MLR and RAS because both remain inside the boundary. What remains is

$$Ax = q_E A c_E + w A c_U = A g_E + \frac{w}{q_U} A g_U. \quad (19)$$

This is the engineer's external check: conserved inventory enters only with fresh wastewater and leaves only with treated overflow and wasted sludge.

2.4. One Identifiable Response for the Complete Loop

Separate unit surrogates would have to be iterated around two recycle paths. Instead, collect every quantity needed to close those paths in one vector:

$$\chi = \begin{bmatrix} m \\ c_1 \\ \vdots \\ c_N \\ g_E \\ g_U \\ s \end{bmatrix} \in \mathbb{R}^L, \quad L = (N + 3)F + N_\ell. \quad (20)$$

For the five-CSTR, 20-component, ten-layer case, $L = 170$. Concentrations, normalized mass flows, and layer TSS have different units, so the regression is built in standardized coordinates. Let

$$\tilde{\vartheta} = S_\vartheta^{-1}(\vartheta - \mu_\vartheta), \quad \tilde{x} = S_x^{-1}(x - \mu_x), \quad \tilde{\chi} = D_\chi^{-1}(\chi - \mu_\chi), \quad (21)$$

where all centers and positive diagonal scales come only from the applicable fitting rows.

A second-order model needs squares and cross-products, but it needs each monomial only once. Let $q(v)$ list $v_j v_k$ for $j \leq k$, and form the nonconstant vector

$$\psi(\vartheta, x) = \begin{bmatrix} \tilde{\vartheta} \\ \tilde{x} \\ q(\tilde{\vartheta}) \\ q(\tilde{x}) \\ \tilde{\vartheta} \otimes \tilde{x} \end{bmatrix}. \quad (22)$$

Centering and scaling these nonconstant terms once more gives the serialized feature vector

$$\phi(\vartheta, x) = \begin{bmatrix} 1 \\ S_\psi^{-1}[\psi(\vartheta, x) - \mu_\psi] \end{bmatrix}, \quad d_\phi = 1 + n_\vartheta + F + \frac{n_\vartheta(n_\vartheta + 1)}{2} + \frac{F(F + 1)}{2} + n_\vartheta F. \quad (23)$$

For $n_\vartheta = 5$ and $F = 20$, the basis contains $d_\phi = 351$ columns. It includes every linear term, square, within-vector cross-product, and design–influent interaction exactly once. Within $q(v)$ the pairs are serialized lexicographically as $(1, 1), (1, 2), \dots, (1, d), (2, 2), \dots, (d, d)$; the Kronecker block is serialized with the influent index varying fastest.

The frozen raw prediction is

$$\tilde{\chi}_{\text{raw}} = \hat{C}\phi(\vartheta, x), \quad \chi_{\text{raw}} = \mu_\chi + D_\chi \tilde{\chi}_{\text{raw}}. \quad (24)$$

The matrix $\hat{C} \in \mathbb{R}^{L \times d_\phi}$ is the complete input–response map. Its design matrix must pass the full-column-rank and condition-number tests stated in the training procedure. For a fixed influent, Equation (24) is a vector-valued quadratic in the design and operating decisions.

2.5. Assembling the Plant Rules into Matrices

The equations developed stream by stream can now be assembled without hiding their physical order. Define the equality residual by

$$\mathcal{H}(\vartheta)\chi - b(x) = \begin{bmatrix} q_P m - r_I c_N - (r_R/q_U)g_U - x \\ A(c_1 - m) \\ A(c_2 - c_1) \\ \vdots \\ A(c_N - c_{N-1}) \\ g_E + g_U - q_C c_N \\ E_S g_U - q_U^\top E_S c_N \\ q_E s_1 - t_{X^\top}^\top g_E \\ q_U s_{N_\ell} - t_{X^\top}^\top g_U \end{bmatrix}. \quad (25)$$

The right-hand side is $b(x) = (x^\top, 0_{R-F}^\top)^\top$. Within every component block the order is Equation (40); within every invariant block the row order is the fixed row order of A . Thus,

$$R = F + NK + F + F_S + 2, \quad (26)$$

where $F_S = 10$ is the number of soluble coordinates. The case has $R = 77$: rows 1–20 are mixer balances, 21–45 are the five stage-invariant blocks, 46–65 are Clarifier component balances, 66–75 are soluble-pass-through balances, and rows 76–77 are the two TSS endpoints.

The particulate and layer relations have the equally explicit order

$$\mathcal{G}(\vartheta)\chi = \begin{bmatrix} q_U E_X c_N - E_X g_U \\ s_1 \mathbf{1}_{N_\ell-2} - (s_2, \dots, s_{N_\ell-1})^\top \\ (s_2, \dots, s_{N_\ell-1})^\top - s_{N_\ell} \mathbf{1}_{N_\ell-2} \end{bmatrix} \leq 0. \quad (27)$$

Here $R_g = F_X + 2(N_\ell - 2) = 26$, with $F_X = 10$. Rows 1–10 follow particulate-component order, rows 11–18 impose the top bounds for layers 2–9, and rows 19–26 impose the bottom bounds in the same layer order. Positive diagonal matrices D_b and D_g , defined from named physical terms in the training section, scale these rows without changing either feasible set.

An equality-only projection is useful when diagnosing how much of a raw error came from broken network closure. With $D_\chi = \text{Diag}(s_{\chi,1}, \dots, s_{\chi,L})$ and every $s_{\chi,j} > 0$, define

$$\chi_{\text{aff}} = \arg \min_{\chi \in \mathbb{R}^L} \frac{1}{2} \|D_\chi^{-1}(\chi - \chi_{\text{raw}})\|_2^2 \quad (28)$$

$$\text{subject to } H(\vartheta)\chi = b(x).$$

Its closed form is

$$\chi_{\text{aff}} = \chi_{\text{raw}} - D_\chi^2 H^\top (H D_\chi^2 H^\top)^{-1} [H \chi_{\text{raw}} - b(x)]. \quad (29)$$

The displayed inverse explains the geometry; the calculation uses linear solves. This diagnostic is evaluated for assessment, but it need not be solved before every optimization QP.

2.6. The Unique Physical Correction

The raw response may break an equality, cross zero, or reverse the expected direction of particulate clarification. To correct all such defects at once and to avoid a poorly scaled Hessian, write a candidate physical state as $\chi = \chi_{\text{raw}} + D_\chi u$, where u is a dimensionless displacement. The deployed state is

$$\begin{aligned} \hat{u} &= \arg \min_{u \in \mathbb{R}^L} \frac{1}{2} \|u\|_2^2 \\ \text{subject to } D_b^{-1} \{H(\vartheta)[\chi_{\text{raw}} + D_\chi u] - b(x)\} &= 0, \\ \chi_{\text{raw}} + D_\chi u &\geq 0, \\ D_g^{-1} \mathcal{G}(\vartheta)[\chi_{\text{raw}} + D_\chi u] &\leq 0. \end{aligned} \quad (30)$$

After the QP is solved, the deployed state is defined by

$$\hat{\chi}(\vartheta; x) = \chi_{\text{raw}} + D_\chi \hat{u}. \quad (31)$$

The identity Hessian makes Equation (30) strictly convex. Its feasibility can be seen without relying on a numerical solver: choose $m = c_1 = \dots = c_N = x$, $g_E = q_E x$, $g_U = q_U x$, and $s_1 = \dots = s_{N_\ell} = t_X^\top x$. This no-conversion point satisfies every equality and inequality. The feasible set is therefore nonempty, and strict convexity gives one unique complete response.

For any equality-feasible χ , weighted Pythagorean decomposition gives

$$\|D_\chi^{-1}(\chi - \chi_{\text{raw}})\|_2^2 = \|D_\chi^{-1}(\chi - \chi_{\text{aff}})\|_2^2 + \text{constant}. \quad (32)$$

Consequently, the direct QP in Equation (30) has the same solution as a QP centered on χ_{aff} , while avoiding a redundant projection during every upper-level evaluation.

The exact deployment contract can now be stated without overstating it. To numerical tolerance, the returned state satisfies

$$\begin{aligned}
 q_P \hat{m} &= x + r_I \hat{c}_N + r_R \hat{c}_U, \\
 A \hat{c}_1 &= A \hat{m}, \quad A \hat{c}_i = A \hat{c}_{i-1}, \quad i = 2, \dots, N, \\
 \hat{g}_E + \hat{g}_U &= q_C \hat{c}_N, \quad E_S \hat{c}_E = E_S \hat{c}_U = E_S \hat{c}_N, \\
 E_X \hat{c}_U &\geq E_X \hat{c}_N, \quad E_X \hat{c}_E \leq E_X \hat{c}_N, \\
 q_E \hat{s}_1 &= t_X^\top \hat{g}_E, \quad q_U \hat{s}_{N_\ell} = t_X^\top \hat{g}_U, \\
 \hat{s}_1 \leq \hat{s}_\ell &\leq \hat{s}_{N_\ell}, \quad \ell = 2, \dots, N_\ell - 1, \quad \hat{\chi} \geq 0.
 \end{aligned} \tag{33}$$

These are network, inventory, and direction guarantees. They do not assert that the projected point satisfies every ASM kinetic equation or every nonlinear Clarifier flux relation. Those stronger claims require the independent mechanistic evaluation described later.

2.7. From a Physical State to an Engineering Decision

The discharge is c_E , not the mixed liquor c_N . Let T select the effluent components or composites to be scored, let D_T contain frozen positive production-response scales, and let $\alpha_Q \geq 0$ with $\mathbf{1}^\top \alpha_Q = 1$. A dimensionless concentration-quality term is

$$J_Q(\chi) = \alpha_Q^\top D_T^{-1} T c_E = \alpha_Q^\top D_T^{-1} T \frac{g_E}{q_E}. \tag{34}$$

Concentration is used because the primary case represents discharge-quality limits; the corresponding loads $Q_0 T g_E$ are retained and reported separately. Let $J_{\text{des/op}}$ contain the declared volume, aeration, pumping, and wasted-solids contributions. The complete engineering score is

$$J(\vartheta, \chi) = \lambda_Q J_Q(\chi) + J_{\text{des/op}}(\vartheta, \chi), \quad \lambda_Q > 0. \tag{35}$$

Equipment and permit limits answer whether a candidate may be operated; they should not move the predicted state merely to manufacture compliance. They are therefore checked after Equation (30). Denote them by $g_{\text{cap}}(\vartheta, \chi) \leq 0$. Whole-plant SRT is one such quantity. Because the Clarifier inventory is derived from the layer profile, it is

$$\text{SRT}(\vartheta, \chi) = \frac{\sum_{i=1}^N V_i t_X^\top c_i + \sum_{\ell=1}^{N_\ell} V_{\text{cl},\ell} s_\ell}{Q_0 (q_E t_X^\top c_E + w t_X^\top c_U)}. \tag{36}$$

The numerator is all suspended solids held in the CSTRs and Clarifier. The denominator contains only solids that cross the plant boundary in effluent and WAS. MLR and RAS are absent because they circulate internally. The SRT is defined only when

$$q_E t_X^\top c_E + w t_X^\top c_U > 10^{-10} \max(1, q_C t_X^\top c_N); \tag{37}$$

otherwise the candidate is rejected instead of being assigned an artificial infinite SRT.

The nested optimization is

$$\begin{aligned}
 \vartheta^* &= \arg \min_{\vartheta} J[\vartheta, \hat{\chi}(\vartheta; x)] \\
 \text{subject to } \vartheta &\in \mathcal{V}, \\
 g_{\text{cap}}[\vartheta, \hat{\chi}(\vartheta; x)] &\leq 0, \\
 \hat{\chi}(\vartheta; x) &\text{ solves Equation (30).}
 \end{aligned} \tag{38}$$

The lower answer is unique for each proposed ϑ . Across proposals, the quadratic surrogate, flow-dependent matrices, recovered concentrations, and changing active inequalities make the upper response nonlinear, generally nonsmooth, and potentially nonconvex.

2.8. Bounded Search and the Meaning of the Reported Answer

Let n_g be the case-study decision dimension and $D_g = \text{Diag}(\vartheta^U - \vartheta^L)$. The outer algorithm works in the unit box through

$$z = D_g^{-1}(\vartheta - \vartheta^L) \in [0, 1]^{n_g}. \quad (39)$$

For each z , the evaluator reconstructs physical decisions, forms the raw state, solves one QP, independently verifies Equation (33), checks capacity constraints, and only then assigns an objective. A failed or infeasible state is recorded and never replaced by the influent, a clipped prediction, or the preceding successful point.

The unit-box center and corners, deterministic full-box and coordinate-face DIRECT exploration (Jones et al., 1993), and bounded multi-basin pattern refinement provide global coverage and explicit boundary checks. Failed and infeasible evaluations receive a finite feasibility-first search merit defined in the methodology. A finite budget does not certify a global optimum; the reported answer is the best verified incumbent at the stated evaluation budget and spatial resolution. The mechanistic reference is interpreted with the same limitation.

3. Methodology

3.1. End-to-End Study Design

The computational study follows four information-separated phases. First, the coupled ASM2d-TSN–Clarifier model generates one fixed 20,000-state design. Second, the fixed estimator is fitted on 16,000 development rows and evaluated once against 4,000 rows held out from that fit using predeclared numerical, predictive, and correction-trust criteria. Third, after that assessment is sealed, the unchanged estimator is refitted on all 20,000 rows and frozen for nominal and robustness optimization. Fourth, the complete mechanistic network is evaluated at every selected point and searched independently. No assessment, nominal, or robustness response alters the mechanistic domain, estimator, features, objective weights, trust limits, or numerical thresholds.

3.2. Coupled Reactor–Clarifier Data Generator

The reaction model is ASM2d with two-step nitrification (ASM2d-TSN), comprising $P = 28$ processes and $F = 20$ components (Guerrero et al., 2011; Henze et al., 2006). Appendix A gives every rate, stoichiometric coefficient, composition coefficient, and parameter value used here. Every matrix, prediction, and physical check uses the fixed order

$$(S_O, S_F, S_A, S_{NH_4}, S_{NO_2}, S_{NO_3}, S_{N_2}, S_{PO_4}, S_I, S_{ALK}, \\ X_I, X_S, X_H, X_{PAO}, X_{PP}, X_{PHA}, X_{AOB}, X_{NOB}, X_{MeP}, X_{MeOH}). \quad (40)$$

The first ten entries are soluble and the last ten particulate. Component units remain those of the mechanistic model; COD, TN, TP, and TSS are calculated only after a complete component state is available through the fixed I_{comp} matrix.

The case study fixes $N = 5$ equal-volume CSTRs. CSTRs 1–2 are unaerated denitrification stages, CSTRs 3–5 share one manipulated aerobic intensity, and MLR is taken from CSTR 5. Appendix A.2 defines a deterministic five-row physical conservation matrix B and its row-normalized form A . For a matrix $C \in \mathbb{R}^{m \times n}$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots$, define $\text{rank}_\epsilon(C)$ as the number satisfying $\sigma_k > \max(m, n)\epsilon_{\text{mach}}\sigma_1$. Generation stops unless $\text{rank}_\epsilon(v) = 15$, $\text{rank}_\epsilon(B) = \text{rank}_\epsilon(A) = 5$, and $\|Av^\top\|_\infty$, $\|Ae_{S_O}\|_\infty$, and $\max_k \||A_{k,:}\|_2 - 1|$ are each no larger than 10^{-10} .

The last CSTR feeds a non-reactive, one-dimensional $N_\ell = 10$ layer Clarifier based on the flux-limited clarification–thickening model of Takács et al. (1991) and the benchmark conventions of Alex et al. (2018) and Jeppsson et al. (2007). To make the numerical generator reproducible, let s_ℓ be layer TSS, $s_F = t_X^\top c_5$ the feed TSS, and

$$v_s(s) = \max\{0, \min[v_0^{\max}, v_0(e^{-r_h(s-s_{\min})} - e^{-r_p(s-s_{\min})})]\}, \quad s_{\min} = f_{ns}s_F. \quad (41)$$

The two exponentials describe hindered settling at high solids and the low-concentration branch; the outer bounds prevent a negative settling speed or a speed above $v_0^{\max}$. Define the unrestricted gravity flux $j_s(s) = v_s(s)s$. At the interface between top-to-bottom layers ℓ and $\ell + 1$, the benchmark receiver-capacity limiter is

$$j_{\ell+1/2}^{\text{set}} = \begin{cases} j_s(s_\ell), & s_{\ell+1} \leq X_t, \\ \min\{j_s(s_\ell), j_s(s_{\ell+1})\}, & s_{\ell+1} > X_t. \end{cases} \quad (42)$$

Table 1

Fixed multilayer-Clarifier geometry and settling parameters.

Quantity	Value	Unit or convention
Surface area A_{cl}	1500	m^2
Total depth	4.0	m
Layer count; depth; volume	10; 0.4; 600	—; m; m^3
Feed layer	5 from top	6 from bottom; array index 4
Maximum settling speed $v_0^{\max}$	250	$m\ d^{-1}$
Theoretical coefficient v_0	474	$m\ d^{-1}$
Hindered coefficient r_h	0.000576	$m^3\ g^{-1}$
Low-concentration coefficient r_p	0.00286	$m^3\ g^{-1}$
Non-settleable fraction f_{ns}	0.00228	dimensionless
Flux-limiting threshold X_l	3000	$g\ TSS\ m^{-3}$
Blanket diagnostic threshold	3000	$g\ TSS\ m^{-3}$

This expression lets dilute receiving layers accept the incoming gravity flux and limits that flux when the receiving layer is already concentrated.

Take downward flux as positive and let the feed enter layer $f = 5$ in a top-to-bottom, one-based array (the fifth layer from the top and sixth from the bottom). Above the feed, bulk liquid moves upward at $v_E = Q_0 q_E / A_{cl}$; below it, bulk liquid moves downward at $v_U = Q_0 q_U / A_{cl}$. The total internal-interface TSS flux is therefore

$$J_{\ell+1/2} = \begin{cases} -v_E s_{\ell+1} + j_{\ell+1/2}^{\text{set}}, & \ell < f, \\ v_U s_{\ell} + j_{\ell+1/2}^{\text{set}}, & \ell \geq f. \end{cases} \quad (43)$$

At the water surface and hopper, $J_{1/2} = -v_E s_1$ and $J_{N_{\ell}+1/2} = v_U s_{N_{\ell}}$; no gravity flux is allowed to cross those physical boundaries. The finite-volume layer balance is

$$V_{cl,\ell} \frac{ds_{\ell}}{dt} = A_{cl}(J_{\ell-1/2} - J_{\ell+1/2}) + \mathbf{1}_{\{\ell=f\}} Q_0 q_C s_F. \quad (44)$$

Summing Equation (44) over all layers cancels internal fluxes and recovers $q_C s_F = q_E s_1 + q_U s_{N_{\ell}}$ at steady state. This is why the endpoint equations in the lower QP are not arbitrary additions.

Soluble components follow the liquid phase unchanged. Particulate components share the TSS flux and retain the Clarifier-feed particulate proportions. The complete layer reconstruction is

$$c_{cl,\ell,j} = \begin{cases} c_{5,j}, & j \in S, \\ (c_{5,j}/s_F) s_{\ell}, & j \in \mathcal{X}, s_F > 0, \\ 0, & j \in \mathcal{X}, s_F = 0, \end{cases} \quad J_{\ell+1/2,j} = (c_{5,j}/s_F) J_{\ell+1/2} \quad (j \in \mathcal{X}, s_F > 0). \quad (45)$$

Thus $c_{E,j} = (c_{5,j}/s_F) s_1$ and $c_{U,j} = (c_{5,j}/s_F) s_{N_{\ell}}$ for particulates. Because every particulate entry of t_X is positive, $s_F = 0$ in a non-negative state implies that all particulate feed coordinates are zero, so the limiting definition is mass consistent.

The mechanistic dynamic state is $y = (c_1, \dots, c_5, s_1, \dots, s_{10}) \in \mathbb{R}_+^{110}$. At every residual evaluation, Equation (45) gives c_E and c_U , and Equation (3) gives m . The transient reactor balance whose steady form is Equation (6) is

$$\frac{dc_i}{dt} = d_i(c_{i-1} - c_i) + v^{\top} \rho(c_i) + \omega_i e_{S_0}, \quad c_0 = m. \quad (46)$$

The five vector derivatives in Equation (46) and the ten scalar layer derivatives in Equation (44) therefore form one coupled function $\dot{y} = f(y; \vartheta, x)$. Eliminating the mixer and outlet algebraic variables in this way does not break the loop: a change in the bottom layer immediately changes RAS, the mixer, CSTR 1, and every downstream state in the same residual evaluation.

The nonlinear residual is expressed in dimensionless-state rates. Reactor component (i, j) is divided by $c_j^{\text{ref}} = \max(1, x_j^U)$, where x_j^U is the upper bound in Table 4; layer derivative ℓ is divided by $s^{\text{ref}} = \max(1, s_F)$. A bounded trust-region-reflective nonlinear least-squares calculation minimizes this scaled derivative vector subject to $y \geq 0$, using $\text{xtol}=\text{ftol}=\text{gtol}=10^{-9}$ and at most 5000 residual evaluations.

Table 2

Mechanistic steady-state numerical and physical acceptance contract.

Check or setting	Limit	Purpose
Nonlinear xtol, ftol, gtol	10^{-9}	Steady nonlinear solve
Maximum residual evaluations	5000 per start	Bounded retry effort
BDF relative; absolute tolerance	10^{-8} ; 10^{-10}	Dynamic fallback
BDF integration horizon	≤ 100 d	Fixed relaxation limit
Scaled CSTR/layer derivative	$\leq 10^{-8}$ d $^{-1}$	Local balance closure
Scaled terminal derivative	$\leq 10^{-8}$ d $^{-1}$	Steady-state confirmation
Minimum state coordinate	$\geq -10^{-10}$	No clipped negative state
Largest real reduced-Jacobian eigenvalue	$\leq 10^{-8}$ d $^{-1}$	Local stability screen
Component Clarifier and plant-boundary residual	$\leq 10^{-8}$ scaled	External closure

Two initial states are prescribed. Start 1 uses $c_i = x$ for every stage and $s_\ell = t_X^\top x$ for every layer. Start 2 uses

$$c_i = E_S^\top E_S x + \alpha_i E_X^\top E_X x, \quad (\alpha_1, \dots, \alpha_5) = (1.5, 2.0, 2.5, 3.0, 3.5), \quad (47)$$

and, after forming $s_F = t_X^\top c_5$,

$$(s_1, \dots, s_{10}) = s_F(0.002, 0.005, 0.01, 0.03, 0.10, 0.50, 1.25, 2.25, 3.25, 4.00). \quad (48)$$

Start 1 is attempted first and fixes the branch whenever it passes the complete acceptance contract. Start 2 is used only when Start 1 fails. If both fail, let y_0 be the returned point with smaller scaled infinity residual. A fixed diagonal integration scale S_y uses $c_j^{\text{ref}} = \max(1, x_j^U)$ for each reactor coordinate and the single value $s_0^{\text{ref}} = \max[1, t_X^\top c_5(y_0)]$ for every layer coordinate. The fallback integrates

$$\frac{d\tilde{y}}{dt} = S_y^{-1} f(S_y \tilde{y}; \vartheta, x), \quad \tilde{y}(0) = S_y^{-1} y_0, \quad (49)$$

with variable-order BDF, relative tolerance 10^{-8} , scaled-coordinate absolute tolerance 10^{-10} , and maximum step 0.25 d. Every Newton update is fraction-to-boundary damped so that $S_y \tilde{y} \geq 0$ before a rate is evaluated; a step that cannot obtain a non-negative converged iterate is rejected and retried at half its step size. No state is clipped. At most 20 consecutive positivity rejections and 100,000 total step attempts are allowed, and no attempted step may be smaller than 10^{-8} d; exceeding any limit makes the fallback fail. Integration stops at 100 d or after the scaled derivative infinity norm is no larger than 10^{-8} d $^{-1}$ at two consecutive accepted endpoints, whichever comes first. That endpoint is supplied once to the same bounded steady solver. No preceding Latin-hypercube solution is used as an initial state.

For the stability screen, the reduced Jacobian is $J_y = \partial f / \partial y$ after the algebraic quantities have been eliminated. Column j is calculated by a centered finite difference with step $\sqrt{\epsilon_{\text{mach}}} \max(1, |y_j|)$, using a forward difference only when the centered step would cross zero. The deployment QP's uniqueness is not interpreted as uniqueness of this nonlinear steady state; the prescribed initialization route defines the branch represented in the data.

For every algebraic balance written as $\sum_k t_k = 0$, its scaled replay residual is $|\sum_k t_k| / \max(1, \sum_k |t_k|)$. The 10^{-8} external-closure limit in Table 2 applies to this quantity. Every accepted state also passes hydraulic splits, RAS/WAS common composition, soluble pass-through, $q_U/q_C - 10^{-10} \leq \eta_j \leq 1 + 10^{-10}$ for every available particulate ratio, the layer envelope, and finite-rate checks. The lower recovery bound is the mechanistic counterpart of $c_{U,j} \geq c_{5,j}$ and makes an accepted target feasible for the deployment inequalities to the stated tolerance. Every ρ_p , oxygen-transfer term, settling velocity, and interface flux must be finite and no smaller than -10^{-12} when its mathematical definition is non-negative. A violation invalidates the state; negative quantities are not clipped into acceptance.

3.3. Case-Study Design, Operating, and Influent Domains

The primary case reduces the general vector in Equation (4) to five decisions:

$$\vartheta = (H, a, r_I, r_R, w)^\top. \quad (50)$$

The total nominal HRT is divided equally, so $\tau_i = H/5$ and $V_i = Q_0 H / (24 \times 5)$. Aeration is zero in CSTRs 1–2 and $k_L a = 47a$ d $^{-1}$ in CSTRs 3–5. Table 3 is both the training and optimization box; no surrogate response outside it is scored.

Table 3

Fixed topology, joint design-and-operation domain, and upper-level screens.

Quantity	Value or range	Interpretation
Fresh flow Q_0	10,000 m ³ d ⁻¹	Plant throughput basis
Number of CSTRs N	5	Fixed topology
Nominal total HRT H	[6, 36] h	Sets five equal tank volumes
Aerobic setting a	[0, 1]	$k_L a = 47a$ d ⁻¹ in CSTRs 3–5
MLR ratio r_I	[0, 4]	Q_I/Q_0
RAS ratio r_R	[0.25, 1.25]	Q_R/Q_0
WAS ratio w	[0.001, 0.05]	Q_W/Q_0
Dissolved-oxygen saturation	8.5 g O ₂ m ⁻³	Aeration boundary
SRT acceptance interval	[8, 30] d	Upper-level capacity constraint
Clarifier surface overflow	≤ 20 m d ⁻¹	Upper-level hydraulic screen
Clarifier feed solids loading	≤ 100 kg TSS m ⁻² d ⁻¹	Upper-level solids screen
Underflow TSS $X_U^{\max}$	≤ 15,000 g TSS m ⁻³	Pumpability screen

Table 4

Fresh-influent component ranges for mechanistic design and surrogate fitting.

Component	Range	Unit	Component	Range	Unit
S_O	[0.0, 0.5]	g O ₂ m ⁻³	S_I	[10, 90]	g COD m ⁻³
S_F	[20, 180]	g COD m ⁻³	S_{ALK}	[1.6, 5.2]	mol HCO ₃ ⁻ m ⁻³
S_A	[5, 80]	g COD m ⁻³	X_I	[20, 120]	g COD m ⁻³
S_{NH4}	[12, 55]	g N m ⁻³	X_S	[60, 280]	g COD m ⁻³
S_{NO2}	[0, 3]	g N m ⁻³	X_H	[15, 100]	g COD m ⁻³
S_{NO3}	[0, 8]	g N m ⁻³	X_{PAO}	[5, 60]	g COD m ⁻³
S_{N2}	[0, 2]	g N m ⁻³	X_{PP}	[2, 20]	g P m ⁻³
S_{PO4}	[2, 18]	g P m ⁻³	X_{PHA}	[1, 30]	g COD m ⁻³
			X_{AOB}	[0.5, 8]	g COD m ⁻³
			X_{NOB}	[0.5, 8]	g COD m ⁻³
			X_{MeP}	[0, 12]	g TSS m ⁻³
			X_{MeOH}	[0, 12]	g TSS m ⁻³

The surface-overflow and solids-loading screens are

$$\text{SOR} = \frac{Q_0 q_E}{A_{cl}}, \quad \text{SLR} = \frac{Q_0 q_C t_X^T c_5}{1000 A_{cl}}. \quad (51)$$

The factor 1000 converts grams to kilograms in SLR. These are case-study engineering screens (Tchobanoglous et al., 2014), not fitted Takács parameters. At fixed Q_0 and area, SOR cannot become active over the declared WAS range but is retained as a hydraulic safety check. The underflow-TSS limit is repeated at 12,000 and 20,000 g TSS m⁻³ without changing the frozen surrogate.

Table 4 completes the 25-dimensional training domain. The alkalinity range is expressed in the ASM unit mol HCO₃⁻ m⁻³: 80–260 g CaCO₃ m⁻³ converts to 1.6–5.2 mol of charge m⁻³, which is the range used in the component vector.

The nominal influent is the componentwise midpoint of Table 4, in the order of Equation (40):

$$x_{\text{nom}} = (0.25, 100, 42.5, 33.5, 1.5, 4, 1, 10, 50, 3.4, 70, 170, 57.5, 32.5, 11, 15.5, 4.25, 4.25, 6, 6)^T. \quad (52)$$

It is a designed in-domain influent, not a measured plant sample. Over the declared box,

$$q_P \in [1.25, 6.25], \quad H/5 \in [1.2, 7.2] \text{ h}, \quad H/(5q_P) \in [0.192, 5.76] \text{ h}. \quad (53)$$

These are derived hydraulic supports of the joint design rather than additional acceptance constraints.

3.4. Mechanistic Design and Stored Targets

A single strength-1 Latin hypercube (McKay et al., 1979) contains exactly $n_D = 20,000$ points in the 25-dimensional box, ordered as $(H, a, r_I, r_R, w, x_1, \dots, x_{20})$. Its random stream is defined here so that no software-specific seeding convention is required. All variables in the following generator are unsigned 64-bit integers; $\oplus$ is bitwise

exclusive-or, $\gg$ is a logical right shift, and every addition and multiplication is reduced modulo 2^{64} . Starting from $q_0 = 42$, draw the 64-bit word $\mathcal{U}_t$ by

$$q_{t+1} = q_t + 9E3779B97F4A7C15_{16}, \quad (54)$$

$$v_1 = [q_{t+1} \oplus (q_{t+1} \gg 30)] BF58476D1CE4E5B9_{16}, \quad (55)$$

$$v_2 = [v_1 \oplus (v_1 \gg 27)] 94D049BB133111EB_{16}, \quad (56)$$

$$\mathcal{U}_t = v_2 \oplus (v_2 \gg 31). \quad (57)$$

The draw counter t advances once for every generated word, including a rejected word. For dimension j , initialize $\pi_j = (0, \dots, n_D - 1)$. For the separate shuffle index $k = n_D - 1, \dots, 1$, let $b = k + 1$ and $L_b = 2^{64} - (2^{64} \bmod b)$; draw until the current $\mathcal{U}_t < L_b$ and swap entries k and $\mathcal{U}_t \bmod b$. This rejection step gives an unbiased Fisher–Yates permutation. After that permutation is complete, draw once for each row in increasing order and set $\xi_{ij} = (\mathcal{U}_t \gg 11)/2^{53}$. The unit-box coordinate is

$$\zeta_{ij} = \frac{\pi_j(i-1) + \xi_{ij}}{n_D}, \quad i = 1, \dots, n_D. \quad (58)$$

Affine range maps send ζ_{ij} to the physical variables. Dimensions and rows are generated in the displayed order, which fixes random-number consumption. Each point must produce an accepted state under Table 2. A failed point is not replaced. If any point remains unresolved after the fixed retry route, the study stops without a fitted response. All start attempts and diagnostics are retained. The robustness design restarts the same generator with the scalar state stated in its subsection.

For each accepted row, the target vector is

$$\chi_{ASM} = (m, c_1, \dots, c_5, g_E, g_U, s_1, \dots, s_{10}) \in \mathbb{R}^{170}. \quad (59)$$

The full component state of every Clarifier layer, c_E , c_U , component and TSS recoveries, composites, loadings, SRT, reaction and oxygen-transfer rates, initialization route, stability result, residuals, and solve time are retained as diagnostics. Outlet mass flows are modeled because their scale remains well behaved when q_U is small; the layer TSS values make M_{cl} a derived quantity through Equation (18).

The 20,000 design rows retain their immutable Latin-hypercube generation order. Rows 1–16,000 form the development set $\mathcal{D}$, and rows 16,001–20,000 form the predictive-assessment set $\mathcal{A}$. This membership is fixed before any response is examined. The assessment set is held out from the development fit and is used once as a pass–fail evaluation; it does not alter the estimator, features, numerical thresholds, or trust criteria. It assesses the fixed procedure as applied to the development fit. Because these rows subsequently enter the production refit, its metrics are not described as holdout errors of the final 20,000-row coefficient array.

3.5. Fixed Standardized Multiresponse Least-Squares Estimator

For a fitting index set $\mathcal{I}$ containing n rows, define the population mean, population standard deviation, and numerical reference magnitude of a scalar coordinate v by

$$\bar{v}_{\mathcal{I}} = \frac{1}{n} \sum_{i \in \mathcal{I}} v_i, \quad s_{\mathcal{I}}(v) = \left[\frac{1}{n} \sum_{i \in \mathcal{I}} (v_i - \bar{v}_{\mathcal{I}})^2 \right]^{1/2}, \quad a_{\mathcal{I}}(v) = \max \left(1, \max_{i \in \mathcal{I}} |v_i| \right). \quad (60)$$

A coordinate is numerically variable only when

$$s_{\mathcal{I}}(v) > 10^{-12} a_{\mathcal{I}}(v). \quad (61)$$

This rule is applied to every design coordinate, influent coordinate, nonconstant monomial, and target coordinate. All 350 nonconstant feature coordinates and all 170 target coordinates must pass; the leading constant feature is retained by definition. Otherwise the fit is invalid.

Using only rows in $\mathcal{I}$, construct all centers and scales in Equations (21)–(23). Let $Y_{\mathcal{I}}$ contain the mechanistic targets from Equation (59), in the same row order as the features. The fitting matrices are

$$\Phi_{\mathcal{I}} = \begin{bmatrix} \phi_1^T \\ \vdots \\ \phi_n^T \end{bmatrix} \in \mathbb{R}^{n \times d_{\phi}}, \quad \tilde{Y}_{\mathcal{I}} = (Y_{\mathcal{I}} - \mathbf{1}_n \mu_{\chi}^T) D_{\chi}^{-1} \in \mathbb{R}^{n \times L}, \quad (62)$$

where $d_\phi = 351$ and $L = 170$. The fixed estimator is

$$\hat{C}_I = \arg \min_{C \in \mathbb{R}^{L \times d_\phi}} \frac{1}{nL} \left\| \tilde{Y}_I - \Phi_I C^\top \right\|_F^2. \quad (63)$$

Every output is estimated from the single fixed standardized design in Equation (62).

A thin singular-value decomposition

$$\Phi_I = U \Sigma V^\top, \quad \sigma_1 \geq \dots \geq \sigma_{d_\phi} \geq 0, \quad (64)$$

defines

$$\tau_{\text{rank}} = \max(n, d_\phi) \varepsilon_{\text{mach}} \sigma_1, \quad \kappa_2(\Phi_I) = \frac{\sigma_1}{\sigma_{d_\phi}}. \quad (65)$$

The design is admissible only when

$$\sigma_{d_\phi} > \tau_{\text{rank}}, \quad \kappa_2(\Phi_I) \leq 10^8. \quad (66)$$

Under these conditions Equation (63) has one unique least-squares minimizer supported by the observed design.

The coefficients are calculated by column-pivoted Householder QR,

$$\Phi_I \Pi = Q R_\Phi, \quad \hat{C}_I^\top = W_{\text{QR}} = \Pi R_\Phi^{-1} Q^\top \tilde{Y}_I. \quad (67)$$

The independent SVD calculation is

$$W_{\text{SVD}} = V \Sigma^{-1} U^\top \tilde{Y}_I. \quad (68)$$

With $R_Y = \Phi_I W_{\text{QR}} - \tilde{Y}_I$, define

$$\eta_{\text{opt}} = \frac{\|\Phi_I^\top R_Y\|_F}{\max\{1, \|\Phi_I\|_2 \|R_Y\|_F\}}, \quad \eta_{\text{agree}} = \frac{\|W_{\text{QR}} - W_{\text{SVD}}\|_F}{\max(1, \|W_{\text{SVD}}\|_F)}, \quad \max(\eta_{\text{opt}}, \eta_{\text{agree}}) \leq 100 \kappa_2(\Phi_I) \varepsilon_{\text{mach}}. \quad (69)$$

All coefficients must also be finite. Failure of Equations (61), (66), or (69) invalidates the response; no alternate estimator is substituted.

The equality and inequality scales are calculated from physical terms, not from the nearly zero residuals of accepted mechanistic states. For fitting row i , define the signed equality term list $\mathcal{T}_{ir}^{(b)}$ as follows: mixer component j uses $\{q_P m_j, -x_j, -r_I c_{N,j}, -(r_R/q_U) g_{U,j}\}$; the first invariant block uses $\{(Ac_1)_k, -(Am)_k\}$; invariant block $h \geq 2$ uses $\{(Ac_h)_k, -(Ac_{h-1})_k\}$; Clarifier component j uses $\{g_{E,j}, g_{U,j}, -q_C c_{N,j}\}$; soluble component j uses $\{(E_S g_U)_j, -q_U (E_S c_N)_j\}$; and the endpoint rows use $\{q_E s_1, -t_X^\top g_E\}$ and $\{q_U s_{N_\ell}, -t_X^\top g_U\}$. Then

$$(D_b)_{rr} = \max \left\{ 10^{-12}, \left[\frac{1}{n} \sum_{i \in I} \frac{1}{|\mathcal{T}_{ir}^{(b)}|} \sum_{t \in \mathcal{T}_{ir}^{(b)}} t^2 \right]^{1/2} \right\}. \quad (70)$$

For inequality row r , the term lists are $\{q_U (E_X c_N)_j, -(E_X g_U)_j\}$, $\{s_1, -s_\ell\}$, or $\{s_\ell, -s_{N_\ell}\}$ in the row order of Equation (27), and

$$(D_g)_{rr} = \max \left\{ 10^{-12}, \left[\frac{1}{n} \sum_{i \in I} \frac{1}{|\mathcal{T}_{ir}^{(g)}|} \sum_{t \in \mathcal{T}_{ir}^{(g)}} t^2 \right]^{1/2} \right\}. \quad (71)$$

The grouping and row order in Equations (70)–(71) are part of the deployed-response and lower-QP definition.

3.6. Computational Feasibility Gate

The feasibility gate is executed before the 4,000-row assessment, production refit, and optimization searches. The fixed estimator requires two factorizations: a $16,000 \times 351$ development design and a $20,000 \times 351$ production design, each with 170 simultaneous right-hand sides. Define the logical scientific workloads

$$N_{QP}^{sci} = 4000 + 20000 + 101(25000) + 101 = 2,549,101, \quad N_{mech}^{sci} = 20000 + 10000 + 100(2500) = 280,000. \quad (72)$$

The final 101 QPs are cold incumbent replays. These counts exclude designated timing repetitions and count an evaluation once even if its prescribed internal retry is used; the end-to-end timing quantities below include that retry route. They are workload projections, not counts of internal linear-system or nonlinear-residual evaluations.

The first 256 Latin-hypercube points form a timing preflight while remaining part of the immutable design. Their complete retry times define $t_{mech,95}$. The remaining mechanistic design is generated only when $N_{mech}^{sci} t_{mech,95}$ fits the declared core-time envelope. After the 16,000-row development fit, 1000 evenly spaced development rows are cold-deployed to define $t_{QP,95}$, and the observed development factorization time is t_{fit} . The factor 2.25 below accounts for the measured development fit and a production fit projected in proportion to row count. The conservative single-thread scientific-work projection is

$$T_{proj} = 2.25 t_{fit} + N_{QP}^{sci} t_{QP,95} + N_{mech}^{sci} t_{mech,95}. \quad (73)$$

If M_{fit} , M_{QP} , and M_{mech} are the measured preflight peak resident memories, set $M_{proj} = \max(1.25 M_{fit}, M_{QP}, M_{mech})$. The study proceeds to assessment only when $T_{proj} \leq 30$ core-days, $M_{proj} \leq 51.2$ GiB within a 64-GiB allocation, every preflight state passes its full acceptance contract, and every preflight QP passes Equation (91). Failure stops the study; timing observations do not change the estimator, tolerances, sample counts, or search budgets. The diagnostic timing repetitions reported later are accounted separately and do not enter these scientific-evaluation counts.

3.7. One-Time Predictive Assessment and Production Refit

The development response uses $\mathcal{I} = \mathcal{D}$. Its centers, scales, coefficient matrix, and row scales remain fixed while all 4,000 assessment rows are evaluated. For assessment row i , coordinate j , and prediction form $h \in \{\text{raw}, \text{aff}, \text{dep}\}$, let $e_{ij}^{(h)} = \hat{y}_{ij}^{(h)} - Y_{ij}$. With $n_A = 4000$,

$$RMSE_j^{(h)} = \left[\frac{1}{n_A} \sum_{i \in \mathcal{A}} (e_{ij}^{(h)})^2 \right]^{1/2}, \quad MAE_j^{(h)} = \frac{1}{n_A} \sum_{i \in \mathcal{A}} |e_{ij}^{(h)}|, \quad (74)$$

$$Bias_j^{(h)} = \frac{1}{n_A} \sum_{i \in \mathcal{A}} e_{ij}^{(h)}, \quad nRMSE_j^{(h)} = \frac{RMSE_j^{(h)}}{s_{\chi,j}}, \quad (75)$$

$$R_j^{2,(h)} = 1 - \frac{\sum_{i \in \mathcal{A}} (e_{ij}^{(h)})^2}{\sum_{i \in \mathcal{A}} (Y_{ij} - \bar{Y}_{\mathcal{A},j})^2}. \quad (76)$$

For the R^2 denominator, Equation (61) is applied to target coordinate j with $\mathcal{I} = \mathcal{A}$; a failing denominator is reported as undefined. For a coordinate block $\mathcal{B}$, the standardized block error is

$$E_B^{(h)} = \left[\frac{1}{n_A |\mathcal{B}|} \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{B}} \left(\frac{e_{ij}^{(h)}}{s_{\chi,j}} \right)^2 \right]^{1/2}. \quad (77)$$

It is reported for the mixer, every CSTR, both outlet-flow blocks, the layer profile, and the complete state. Overflow and underflow COD, TN, TP, and TSS and normalized Clarifier inventory use the same RMSE, MAE, bias, nRMSE, and R^2 definitions. Each derived-response standard deviation is calculated on $\mathcal{D}$ and must pass Equation (61) before it is used as an nRMSE denominator; failure invalidates the assessment.

The assessment also quantifies reliance on projection. Let

$$e_i^{\text{raw}} = D_{\chi}^{-1}(\chi_{\text{raw},i} - Y_i), \quad e_i^{\text{dep}} = D_{\chi}^{-1}(\hat{\chi}_i - Y_i), \quad u_i = D_{\chi}^{-1}(\hat{\chi}_i - \chi_{\text{raw},i}), \quad \kappa_{\text{corr},i} = \frac{\|u_i\|_2}{\sqrt{L}}. \quad (78)$$

Define

$$E_{\text{state}}^{\text{dep}} = \left[\frac{1}{n_A L} \sum_{i \in \mathcal{A}} \|e_i^{\text{dep}}\|_2^2 \right]^{1/2}, \quad \rho_{\text{corr}} = \left[\frac{\sum_{i \in \mathcal{A}} \|u_i\|_2^2}{\max(10^{-24}, \sum_{i \in \mathcal{A}} \|e_i^{\text{raw}}\|_2^2)} \right]^{1/2}. \quad (79)$$

If $\kappa_{\text{corr},(1)} \leq \dots \leq \kappa_{\text{corr},(n_A)}$ are ordered, then $Q_{0.95}(\kappa_{\text{corr}}) = \kappa_{\text{corr},([0.95n_A])}$.

The development response is admissible only when all 4,000 deployed QPs pass Equation (91), $E_{\text{state}}^{\text{dep}} < 1$, $\rho_{\text{corr}} \leq 0.50$, $Q_{0.95}(\kappa_{\text{corr}}) \leq 0.50$, and

$$\|e_i^{\text{dep}}\|_2^2 \leq \|e_i^{\text{raw}}\|_2^2 + 10^{-6} (1 + \|e_i^{\text{raw}}\|_2^2), \quad i \in \mathcal{A}. \quad (80)$$

The final inequality checks the non-expansive geometry of projection relative to an accepted mechanistic state. Failure of any assessment condition ends the study; assessment responses do not alter the estimator.

After the assessment record is sealed, the production response is refitted on all 20,000 rows using the unchanged equations. Every center, scale, row scale, QR factorization, SVD check, and coefficient is recomputed. The four mechanistic overflow composite coordinates used in D_T must each pass Equation (61) on the 20,000 production rows before their population standard deviations are inverted. The production response is frozen only after Equations (61), (66), and (69) pass and all 20,000 production-row deployments pass Equation (91).

3.8. Nominal Objective and Constraints

For Equation (52), set $T = I_{\text{comp}}$ and $\alpha_Q = \frac{1}{4} \mathbf{1}_4$. Let D_T contain production-fit standard deviations of overflow COD, TN, TP, and TSS. The primary objective is

$$J = 0.50 \alpha_Q^\top D_T^{-1} I_{\text{comp}} c_E + 0.15 \frac{H - H^L}{H^U - H^L} + 0.20 a + 0.05 \frac{r_I - r_I^L}{r_I^U - r_I^L} + 0.05 \frac{r_R - r_R^L}{r_R^U - r_R^L} + 0.05 \frac{w t_X^\top c_U}{w^U X_U^{\max}}. \quad (81)$$

The final term is proportional to the wasted-solids mass rate $Q_0 w t_X^\top c_U$; penalizing w alone would treat a dilute and a concentrated waste stream as equally costly. The constant Q_0 cancels in the normalization. All six coefficients are declared before optimization and sum to one. They define a transparent dimensionless case, not a claim of universal economic equivalence. Sensitivity runs change the five non-quality coefficients while retaining the frozen surrogate and original normalization constants.

The production design also defines a leverage support screen. With Φ_P denoting the 20,000-row production feature matrix,

$$\ell(\vartheta, x) = \phi(\vartheta, x)^\top (\Phi_P^\top \Phi_P)^{-1} \phi(\vartheta, x), \quad \ell_{\max} = \max_{i=1, \dots, 20000} \ell(\vartheta_i, x_i). \quad (82)$$

The displayed inverse defines the scalar; triangular solves with the production QR factor are used numerically.

Three further quantities guard the physical relaxation. The correction magnitude is $\kappa_{\text{corr}} = \|\hat{u}\|_2 / \sqrt{L}$. For particulate coordinates whose feed mass exceeds $10^{-10} q_C \max(1, |\mu_{c_{N,j}}| + s_{c_{N,j}})$, let $\Delta_\eta = \max_j \eta_j - \min_j \eta_j$; set $\Delta_\eta = 0$ if fewer than two ratios are available. Finally, calculate the layer fluxes in Equations (42)–(43) from the deployed profile and define

$$r_{\text{flux}} = \frac{\max_{\ell} |A_{\text{cl}}(J_{\ell-1/2} - J_{\ell+1/2}) + \mathbf{1}_{\{\ell=f\}} Q_0 q_C s_F|}{Q_0 \max(q_C s_F, 1)}. \quad (83)$$

Thus a candidate cannot obtain a favorable SRT by relying on a large projection, unsupported polynomial leverage, component-selective particulate capture, or a layer profile that strongly violates the nonlinear Clarifier flux equation.

No additional discharge constraint is active in the primary case; COD, TN, TP, and TSS are represented in the objective. The complete dimensionless upper-constraint vector is

$$\tilde{g}_{\text{cap}} = \begin{bmatrix} (8 - \text{SRT})/8 \\ (\text{SRT} - 30)/30 \\ (\text{SOR} - 20)/20 \\ (\text{SLR} - 100)/100 \\ (t_X^T c_U - 15000)/15000 \\ (\kappa_{\text{corr}} - 0.50)/0.50 \\ (\ell - \ell_{\text{max}})/\max(\ell_{\text{max}}, 10^{-12}) \\ (\Delta_\eta - 0.05)/0.05 \\ (r_{\text{flux}} - 0.05)/0.05 \end{bmatrix}. \quad (84)$$

Here the dimensional limits use days, m d^{-1} , $\text{kg TSS m}^{-2} \text{d}^{-1}$, and g TSS m^{-3} as appropriate. A successfully deployed candidate is upper-feasible when $\|\tilde{g}_{\text{cap}}\|_\infty \leq 10^{-8}$. These requirements are checked after the lower QP and therefore cannot move its state toward compliance.

3.9. Lower-Level Numerical Acceptance

For a fixed candidate, put Equation (30) in the standard scaled form

$$E = D_b^{-1} H D_\chi, \quad e = D_b^{-1} (b - H \chi_{\text{raw}}), \quad I = \begin{bmatrix} -I_L \\ D_g^{-1} \mathcal{G} D_\chi \end{bmatrix}, \quad h_{\text{QP}} = \begin{bmatrix} D_\chi^{-1} \chi_{\text{raw}} \\ -D_g^{-1} \mathcal{G} \chi_{\text{raw}} \end{bmatrix}. \quad (85)$$

Thus $\hat{u}$ minimizes $\frac{1}{2} u^T u$ subject to $Eu = e$ and $Iu \leq h_{\text{QP}}$. If $\sigma_1 \geq \dots \geq \sigma_R$ are the singular values of E , the candidate is rejected unless

$$\sigma_R > \max(R, L) \epsilon_{\text{mach}} \sigma_1. \quad (86)$$

Let $\lambda_E \in \mathbb{R}^R$ be the equality multiplier, $\lambda_I \in \mathbb{R}_+^{L+R_g}$ the inequality multiplier, and $\varsigma = h_{\text{QP}} - I\hat{u}$ the inequality slack. The independently recomputed KKT relations are

$$E\hat{u} = e, \quad \varsigma \geq 0, \quad \lambda_I \geq 0, \quad \hat{u} + E^T \lambda_E + I^T \lambda_I = 0, \quad \lambda_I \odot \varsigma = 0. \quad (87)$$

Their dimensionless residuals are

$$r_E = \|E\hat{u} - e\|_\infty, \quad r_G = \|[D_g^{-1} \mathcal{G} \hat{\chi}]_+\|_\infty, \quad (88)$$

$$r_+ = \|[-D_\chi^{-1} \hat{\chi}]_+\|_\infty, \quad r_{\text{dual}} = \frac{\|[-\lambda_I]_+\|_\infty}{1 + \|\lambda_I\|_\infty}, \quad (89)$$

$$r_{\text{stat}} = \frac{\|\hat{u} + E^T \lambda_E + I^T \lambda_I\|_\infty}{1 + \|\hat{u}\|_\infty + \|E^T \lambda_E\|_\infty + \|I^T \lambda_I\|_\infty}, \quad r_{\text{comp}} = \frac{\|\lambda_I \odot \varsigma\|_\infty}{1 + \|\lambda_I\|_\infty \|\varsigma\|_\infty}. \quad (90)$$

Acceptance requires finite primal and dual arrays and

$$r_E, r_G, r_{\text{stat}}, r_{\text{dual}}, r_{\text{comp}} \leq 10^{-8}, \quad r_+ \leq 10^{-10}. \quad (91)$$

Multipliers in these equations correspond to the displayed scaled matrices regardless of internal solver scaling. The QP uses double precision, absolute and relative tolerances 10^{-8} , polishing, and at most 100,000 iterations (Stellato et al., 2018). Any solver status is provisional until Equation (91) passes. Assessment rows, production replays, search centers and corners, timing pilots, and final incumbent replays start from zero primal and dual arrays. A DIRECT trial warm-starts only from its parent rectangle center when that center has an accepted lower solution; a pattern trial warm-starts only from its poll center. Trials in one synchronous batch never warm-start one another. When the prescribed warm solve fails, one cold retry reconstructs the factorization and starts all primal and dual variables at zero. A failed retry invalidates the candidate. The plant-boundary identity, Clarifier balances, inventory, and recovered concentrations are also replayed from the unrounded state; no raw, affine, clipped, influent, or preceding state is substituted.

3.10. Outer Search and Mechanistic Validation

The first five entries of Equation (84) are engineering constraints; the final four are surrogate-trust constraints. For a successfully deployed surrogate state, let

$$V_{\text{surr}} = \|\tilde{g}_{\text{cap}}\|_{\infty}. \quad (92)$$

The finite scalar supplied to the search algorithm is

$$F_{\text{search}} = \begin{cases} J/(1+J), & \text{accepted lower state and } V_{\text{surr}} \leq 10^{-8}, \\ 1 + V_{\text{surr}}/(1 + V_{\text{surr}}), & \text{accepted lower state and } V_{\text{surr}} > 10^{-8}, \\ 2, & \text{failed, non-finite, or undefined candidate.} \end{cases} \quad (93)$$

Because $J \geq 0$, every feasible point ranks ahead of every infeasible point, every evaluated infeasible point ranks ahead of a failed point, and ordering by J is unchanged among feasible points. The displayed transforms are part of the optimizer definition: their numerical spacing, not only their ordering, determines the DIRECT lower hull. A failed lower solve, invalid SRT denominator, or non-finite response consumes one distinct-evaluation unit. Candidate keys are the exact binary representations of the normalized coordinates; a cached duplicate consumes no additional unit.

The five controls are normalized to $[0, 1]^5$. The center initializes the full-box DIRECT partition, and all 32 corners are evaluated as separate boundary probes. Corner and later face evaluations are eligible incumbents but do not alter the full-box DIRECT hull. For rectangle r , let c_r be its center, $F_r = F_{\text{search}}(c_r)$, and d_r the Euclidean distance from its center to a vertex. With $F_{\min}$ the smallest merit among centers in the current partition, rectangle r is potentially optimal when some $K_r \geq 0$ satisfies

$$F_r - K_r d_r \leq F_j - K_r d_j \quad \text{for every rectangle } j, \quad F_r - K_r d_r \leq F_{\min} - \epsilon_{\text{DIR}} |F_{\min}|, \quad (94)$$

where $\epsilon_{\text{DIR}} = 10^{-4}$. These inequalities are tested from the lower convex hull of the distinct (d_r, F_r) pairs. A selected rectangle is trisected along every longest coordinate. If its longest side is L_r , the new points are $c_r \pm (L_r/3)e_j$ for those coordinates; coordinates are split in increasing order of the smaller merit at their two trial points, with coordinate index breaking ties. For an exact binary64 tie group with the same (d_r, F_r) , at most one member is split per iteration; iteration number cycles through the members in lexicographic lower-corner order to avoid a fixed spatial bias on flat failure plateaus. Trial batches are evaluated synchronously and committed in lexicographic coordinate order.

The surrogate search has 25,000 distinct-attempt units. The full-box partition contributes at most 18,000 globally new keys, including its center but excluding the 32 separately counted corners, or stops when every rectangle has maximum side length at most $1/1024$. Next, ten four-dimensional DIRECT partitions search the coordinate faces in the order $z_1 = 0, z_1 = 1, \dots, z_5 = 0, z_5 = 1$. Each face contributes at most 300 globally new keys under the same rule and resolution; cached points are free, and a face's hull uses only centers from that face partition. Thus at least 3968 attempts, plus any unused DIRECT quota, remain for local work.

Local seeds are chosen by increasing F_{search} . The best seed is retained, and up to three further seeds are selected greedily subject to ℓ_{∞} separation of at least $1/8$; if necessary, that threshold is halved until four seeds or all distinct basins are obtained. Each seed starts with mesh $\Delta = 1/16$. In cyclic seed order, a complete poll evaluates the axial directions $\{\pm e_j\}$ first and then, lexicographically, all pairwise directions $\{\pm e_j \pm e_k : j < k\}$. Out-of-box coordinates are clipped and duplicates removed. The smallest-merit poll point is accepted if it strictly improves the center; otherwise that seed's mesh is halved. A seed terminates after an unsuccessful poll at $\Delta = 1/512$. A split or poll begins only when all of its uncached trial points fit the phase and total budgets. The final feasible incumbent is recomputed by a cold lower solve. If no feasible point is found, the outcome is "no feasible incumbent"; the smallest violation is reported, but no selected decision or finite regret is defined. This is a finite-resolution refinement, not a stationarity or global-optimality certificate.

For the nominal case and each validation case, the complete mechanistic model is next evaluated at the selected decisions without refitting. Four outcomes are distinguished. If the surrogate search has no feasible incumbent, the case is a surrogate-optimization failure. If the fixed mechanistic solve does not return an accepted, stable steady state, the case is a mechanistic-evaluation failure. If an accepted mechanistic state violates any of the five engineering constraints, the case is a decision-feasibility failure. None of these three outcomes has an ordinary finite regret. If the selected point is accepted and engineering-feasible, it is inserted first into the mechanistic archive and counts against the reference budget.

The mechanistic reference uses the same search merit with V_{ASM} formed from only the five engineering constraints. Its nominal budget is 10,000 distinct attempts and each robustness budget is 2500. Before DIRECT begins, the selected point and the clipped, deduplicated 50-direction pattern stencil at step 2^{-7} are evaluated in center, axial, then lexicographic pairwise order. Corners not already cached are evaluated next. In the nominal reference, the full-box DIRECT partition contributes at most 7000 globally new keys and each coordinate face at most 100; in a robustness reference, the corresponding caps are 1700 and 25. Full-box, corner, stencil, face, failed, and local attempts all count against the stated total. Every remaining unit is assigned to the same multi-basin pattern rule, using up to four nominal seeds and one robustness seed. Even when every preceding quota is filled without a duplicate, at least 1917 nominal units and 467 robustness units remain for local refinement. Every failed mechanistic attempt consumes one unit. If $\hat{\theta}_s$ is an accepted engineering-feasible surrogate decision and $\theta_{\text{ASM},s}^*$ is the best verified mechanistic incumbent, define

$$\mathcal{R}_s = J[\hat{\theta}_s, \chi_{\text{ASM}}(\hat{\theta}_s; x_s)] - J[\theta_{\text{ASM},s}^*, \chi_{\text{ASM}}(\theta_{\text{ASM},s}^*; x_s)]. \quad (95)$$

Because the archive already contains the feasible selected point, its best objective cannot be larger than that point's objective. Hence $\mathcal{R}_s \geq 0$ up to roundoff. A value below -10^{-8} indicates inconsistent objectives, constraints, or archive insertion and invalidates the case. The reported quantity is a gap to a finite-budget incumbent and is not a certificate relative to an unknown global optimum.

3.11. Robustness, Reporting, and Reproducibility

After the nominal problem and all scales are frozen, 100 fresh-influent profiles are generated by the Latin-hypercube construction in the Mechanistic Design subsection, restricted to the 20 influent dimensions and restarting its 64-bit state at $q_0 = 314159$. No response from these profiles enters estimation, scaling, assessment, objective definition, trust limits, or numerical tolerances. Each case retains the selected controls, objective decomposition, raw and corrected states, Clarifier mass flows, recovery and densification factors, physical residuals, mechanistic response, mechanistic reference, outcome class, regret when defined, evaluation counts, and wall-clock durations.

Robustness summaries report means, medians, interquartile ranges, 95th percentiles, maxima, failure frequencies, and bound-activity fractions. Clarifier reporting includes outlet component and composite errors, common-recovery deviation, layer-profile and inventory errors, loadings, SRT, and mass closure. Reactor reporting includes every stage and the axial profiles of dissolved oxygen, nitrogen species, phosphorus species, and solids.

Timing uses a monotonic high-resolution counter with one numerical thread. Separate records are kept for mechanistic design generation, the 16,000-row development fit, assessment, the 20,000-row production fit, raw inference, matrix construction, the lower QP, end-to-end deployed inference, surrogate optimization, selected-point mechanistic simulation, and mechanistic search. Single-state timing uses 100 untimed warmups followed by 1000 fixed-sequence evaluations; batch timing uses two untimed batches followed by 20 batches of 1000. Cold and warm QP setup and solve times remain separate.

The permanent study record contains the random-generator states, component and feature orders, topology, all parameter values, numerical tolerances, fitted coefficients, scales, assessment values, and complete candidate archives. Terminal acceptance replays the 20,000 design rows, 4,000 assessment predictions, nominal case, 100 robustness cases, and every stated physical check. Only a complete record satisfying the contracts in this article is eligible for reporting.

4. Results and Discussion

Three properties follow directly from the formulation. First, the flow definitions close exactly and lead to the plant-boundary inventory in Equation (19); internal recycle ratios change internal concentrations without appearing as external generation or loss. Second, Equation (30) has a nonempty feasible set and an identity Hessian, so every accepted upper candidate has one deployed state. Third, Equation (18) derives Clarifier inventory from ten corrected layer concentrations.

The regression has a direct numerical interpretation. Its 351 dimensionless monomials form one checked full-column-rank design, and a pivoted QR factorization estimates all 170 responses. An independent SVD verifies rank, condition number, and coefficient agreement. The set held out from the development fit quantifies native and standardized prediction errors and measures reliance on physical correction. Production use begins only after the fixed assessment gate passes.

The QP preserves the stated balances, signs, clarification directions, and layer envelope; it does not enforce all 28 reaction-rate equations or the nonlinear flux limiter. Exact equality of every particulate recovery is bilinear when both

feed state and recovery are variables. The leverage, correction, recovery-spread, and flux-residual screens bound this relaxation, while matched mechanistic simulations provide the independent test of kinetics and decision feasibility.

The interpretation of the optimization is similarly bounded. Changing H at fixed Q_0 changes tank volume, so the primary problem is joint design and operation. The number of stages is prescribed, not optimized. Finally, boundary-augmented DIRECT and multi-basin pattern refinement return a verified finite-budget incumbent. Prediction error at the selected point and the non-negative incumbent gap in Equation (95) answer different questions and must be reported separately.

5. Conclusion

The recycling N -stage activated sludge system is represented as one mixer–reactor–Clarifier response and one unique scaled physical projection. Ten Clarifier TSS states make whole-plant SRT traceable, while component mass-flow coordinates describe clarification as recovery into a smaller underflow.

Inputs, targets, and unique quadratic features are standardized from the applicable fitting rows. A full-rank multiresponse least-squares problem is solved by pivoted QR and checked independently by SVD; a set held out from the development fit gates production use; and the unchanged estimator is refitted on all 20,000 states. The physical guarantees are exact for the declared linear network contract, while nonlinear trust screens and independent ASM2d-TSN–Clarifier evaluation address kinetics, flux behavior, operational feasibility, and decision quality. The complete route from process equations to the bounded incumbent is specified in this article.

A. Complete ASM2d-TSN Mechanistic Definition

A.1. Rate functions and process rates

For a non-negative concentration z and a positive half-saturation constant K , define

$$M(z; K) = \frac{z}{K + z}, \quad \overline{M}(z; K) = \frac{K}{K + z}. \quad (96)$$

Let $S_{\text{NOx}} = S_{\text{NO2}} + S_{\text{NO3}}$ and define the electron-acceptor and carbon shares

$$\alpha_2 = \frac{S_{\text{NO2}}}{S_{\text{NOx}}}, \quad \alpha_3 = \frac{S_{\text{NO3}}}{S_{\text{NOx}}}, \quad \alpha_F = \frac{S_F}{S_F + S_A}, \quad \alpha_A = \frac{S_A}{S_F + S_A}. \quad (97)$$

A share is zero when its denominator is zero. Hydrolysable substrate availability and the PAO storage ratios are

$$\theta_X = \frac{X_S}{K_X X_H + X_S}, \quad r_{PP} = \frac{X_{PP}}{X_{PAO}}, \quad r_{PHA} = \frac{X_{PHA}}{X_{PAO}}, \quad (98)$$

The value of θ_X is zero when both X_H and X_S are zero. When $X_{PAO} = 0$, both storage ratios are defined as zero; this convention makes every PAO rate evaluable at the non-negative boundary.

$$\pi_{PP} = M(r_{PP}; K_{PP}), \quad \pi_{PHA} = M(r_{PHA}; K_{PHA}), \quad c_{PP} = \frac{[K_{\max} - r_{PP}]_+}{K_{IPP} + [K_{\max} - r_{PP}]_+}. \quad (99)$$

All PAO rates that multiply X_{PAO} are defined as zero when $X_{PAO} = 0$. The nutrient factors used repeatedly below are

$$L_H = M(S_{NH4}; K_{NH4,H})M(S_{PO4}; K_{PO4,H})M(S_{ALK}; K_{ALK,H}), \quad (100)$$

$$L_P = M(S_{NH4}; K_{NH4,PAO})M(S_{PO4}; K_{PO4,PAO})M(S_{ALK}; K_{ALK,PAO}), \quad (101)$$

$$L_N = M(S_{PO4}; K_{PO4,nit})M(S_{ALK}; K_{ALK,nit}). \quad (102)$$

The hydrolysis, heterotrophic growth, fermentation, and heterotroph-decay rates are

$$\rho_1 = K_H M(S_O; K_{O,hyd}) \theta_X X_H, \quad (103)$$

$$\rho_2 = \eta_{hyd,NO2} K_H \overline{M}(S_O; K_{O,hyd}) M(S_{NO2}; K_{NO2,hyd}) \alpha_2 \theta_X X_H, \quad (104)$$

$$\rho_3 = \eta_{hyd,NO3} K_H \overline{M}(S_O; K_{O,hyd}) M(S_{NO3}; K_{NO3,hyd}) \alpha_3 \theta_X X_H, \quad (105)$$

$$\rho_4 = \eta_{hyd,fe} K_H \overline{M}(S_O; K_{O,hyd}) \overline{M}(S_{NOx}; K_{NOx,hyd}) \theta_X X_H, \quad (106)$$

$$\rho_5 = \mu_H M(S_O; K_{O,H}) M(S_F; K_F) \alpha_F L_H X_H, \quad (107)$$

$$\rho_6 = \mu_H M(S_O; K_{O,H}) M(S_A; K_A) \alpha_A L_H X_H, \quad (108)$$

$$\rho_7 = \mu_H \overline{M}(S_O; K_{O,H}) M(S_F; K_F) \alpha_F L_H \eta_{H,NO3} M(S_{NO3}; K_{NO3,H}) \alpha_3 X_H, \quad (109)$$

$$\rho_8 = \mu_H \overline{M}(S_O; K_{O,H}) M(S_F; K_F) \alpha_F L_H \eta_{H,NO2} M(S_{NO2}; K_{NO2,H}) \alpha_2 X_H, \quad (110)$$

$$\rho_9 = \mu_H \overline{M}(S_O; K_{O,H}) M(S_A; K_A) \alpha_A L_H \eta_{H,NO3} M(S_{NO3}; K_{NO3,H}) \alpha_3 X_H, \quad (111)$$

$$\rho_{10} = \mu_H \overline{M}(S_O; K_{O,H}) M(S_A; K_A) \alpha_A L_H \eta_{H,NO2} M(S_{NO2}; K_{NO2,H}) \alpha_2 X_H, \quad (112)$$

$$\rho_{11} = q_{fe} \overline{M}(S_O; K_{O,H}) \overline{M}(S_{NOx}; K_{NOx,H}) M(S_F; K_{fe}) M(S_{ALK}; K_{ALK,H}) X_H, \quad (113)$$

$$\rho_{12} = b_H X_H. \quad (114)$$

Here q_{fe} enters ρ_{11} directly as a rate constant with units d^{-1} .

The PAO storage, growth, and decay rates are

$$\rho_{13} = q_{PHA} M(S_A; K_A) \overline{M}(S_O; K_{O,PAO}) \overline{M}(S_{NOx}; K_{NOx,PAO}) M(S_{ALK}; K_{ALK,PAO}) \pi_{PP} X_{PAO}, \quad (115)$$

$$\rho_{14} = q_{PP} M(S_O; K_{O,PAO}) M(S_{PO4}; K_{PS}) M(S_{ALK}; K_{ALK,PAO}) \pi_{PHA} c_{PP} X_{PAO}, \quad (116)$$

$$\rho_{15} = q_{PP} \overline{M}(S_O; K_{O,PAO}) M(S_{PO4}; K_{PS}) M(S_{ALK}; K_{ALK,PAO}) \pi_{PHA} c_{PP} \quad (117)$$

$$\times \eta_{PAO,NO3} M(S_{NO3}; K_{NO3,PAO}) \alpha_3 X_{PAO}, \quad (118)$$

$$\rho_{16} = q_{PP} \overline{M}(S_O; K_{O,PAO}) M(S_{PO4}; K_{PS}) M(S_{ALK}; K_{ALK,PAO}) \pi_{PHA} c_{PP} \quad (119)$$

$$\times \eta_{PAO,NO2} M(S_{NO2}; K_{NO2,PAO}) \alpha_2 X_{PAO}, \quad (120)$$

$$\rho_{17} = \mu_{PAO} M(S_O; K_{O,PAO}) L_P \pi_{PHA} X_{PAO}, \quad (121)$$

$$\rho_{18} = \mu_{PAO} \overline{M}(S_O; K_{O,PAO}) L_P \pi_{PHA} \eta_{PAO,NO3} M(S_{NO3}; K_{NO3,PAO}) \alpha_3 X_{PAO}, \quad (122)$$

$$\rho_{19} = \mu_{PAO} \overline{M}(S_O; K_{O,PAO}) L_P \pi_{PHA} \eta_{PAO,NO2} M(S_{NO2}; K_{NO2,PAO}) \alpha_2 X_{PAO}, \quad (123)$$

$$\rho_{20} = b_{PAO} M(S_{ALK}; K_{ALK,PAO}) X_{PAO}, \quad (124)$$

$$\rho_{21} = b_{PP} M(S_{ALK}; K_{ALK,PAO}) X_{PP}, \quad (125)$$

$$\rho_{22} = b_{PHA} M(S_{ALK}; K_{ALK,PAO}) X_{PHA}. \quad (126)$$

The oxygen and NOx inhibition in ρ_{13} makes PHA storage anaerobic. The nitrifier and chemical rates are

$$\rho_{23} = \mu_{AOB} M(S_O; K_{O,AOB}) M(S_{NH4}; K_{NH4,AOB}) L_N X_{AOB}, \quad (127)$$

$$\rho_{24} = \mu_{NOB} M(S_O; K_{O,NOB}) M(S_{NO2}; K_{NO2,NOB}) L_N X_{NOB}, \quad (128)$$

$$\rho_{25} = b_{AOB} X_{AOB}, \quad \rho_{26} = b_{NOB} X_{NOB}, \quad (129)$$

$$\rho_{27} = k_{PRE} S_{PO4} X_{MeOH}, \quad (130)$$

$$\rho_{28} = k_{RED} i_{PM_eP} M(S_{ALK}; K_{ALK,chem}) X_{MeP}. \quad (131)$$

The factor $i_{PM_eP} = 1/4.87 \text{ g P g TSS}^{-1}$ converts redissolving X_{MeP} to the phosphorus basis used by process 28. At a component state c , the process-rate vector used in the reactor equations is

$$\varphi(c) = [\rho_1(c), \dots, \rho_{28}(c)]^\top \in \mathbb{R}_+^{28}, \quad \varphi_i = \varphi(c_i). \quad (132)$$

Rates $\rho_{1:13}$, $\rho_{17:20}$, $\rho_{22:26}$ have a COD mass basis; $\rho_{14:16}$, ρ_{21} , and $\rho_{27:28}$ have a phosphorus mass basis. Each row of v below converts its rate basis into the units of all 20 component coordinates, so $v^\top \varphi(c)$ is a component-rate vector. Oxygen transfer in reactor i is

$$\omega_i = k_L a_i (S_{O,sat} - c_{i,S_O}), \quad S_{O,sat} = 8.5 \text{ g O}_2 \text{ m}^{-3}, \quad (133)$$

with $k_L a_i = 0 \text{ d}^{-1}$ for $i = 1, 2$ and $k_L a_i = 47a \text{ d}^{-1}$ for $i = 3, 4, 5$.

A.2. Stoichiometric matrix and invariant operator

Let $e_j \in \mathbb{R}^F$ be the column coordinate vector for component j ; its transpose is the corresponding row selector. Omitted coefficients below are zero. Define the preliminary process rows

$$\bar{v}_{1:4} = (1 - f_{SI})e_{S_F}^\top + f_{SI}e_{S_I}^\top - e_{X_S}^\top, \quad (134)$$

$$\bar{v}_5 = -Y_H^{-1}e_{S_F}^\top + (1 - Y_H^{-1})e_{S_O}^\top + e_{X_H}^\top, \quad (135)$$

$$\bar{v}_6 = -Y_H^{-1}e_{S_A}^\top + (1 - Y_H^{-1})e_{S_O}^\top + e_{X_H}^\top, \quad (136)$$

$$\bar{v}_7 = -Y_H^{-1}e_{S_F}^\top + \beta_H(e_{S_{NO2}} - e_{S_{NO3}})^\top + e_{X_H}^\top, \quad (137)$$

$$\bar{v}_8 = -Y_H^{-1}e_{S_F}^\top + \gamma_H(e_{S_{N2}} - e_{S_{NO2}})^\top + e_{X_H}^\top, \quad (138)$$

$$\bar{v}_9 = -Y_H^{-1}e_{S_A}^\top + \beta_H(e_{S_{NO2}} - e_{S_{NO3}})^\top + e_{X_H}^\top, \quad (139)$$

$$\bar{v}_{10} = -Y_H^{-1}e_{S_A}^\top + \gamma_H(e_{S_{N2}} - e_{S_{NO2}})^\top + e_{X_H}^\top, \quad (140)$$

$$\bar{v}_{11} = e_{S_A}^\top - e_{S_F}^\top, \quad (141)$$

$$\bar{v}_{12} = f_{XI}e_{X_I}^\top + (1 - f_{XI})e_{X_S}^\top - e_{X_H}^\top, \quad (142)$$

$$\bar{v}_{13} = -e_{S_A}^\top - Y_{PO4}e_{X_{PP}}^\top + e_{X_{PHA}}^\top, \quad (143)$$

$$\bar{v}_{14} = -Y_{PHA}e_{S_O}^\top + e_{X_{PP}}^\top - Y_{PHA}e_{X_{PHA}}^\top, \quad (144)$$

$$\bar{v}_{15} = \beta_S(e_{S_{NO2}} - e_{S_{NO3}})^\top + e_{X_{PP}}^\top - Y_{PHA}e_{X_{PHA}}^\top, \quad (145)$$

$$\bar{v}_{16} = \gamma_S(e_{S_{N2}} - e_{S_{NO2}})^\top + e_{X_{PP}}^\top - Y_{PHA}e_{X_{PHA}}^\top, \quad (146)$$

$$\bar{v}_{17} = (1 - Y_{PAO}^{-1})e_{S_O}^\top + e_{X_{PAO}}^\top - Y_{PAO}^{-1}e_{X_{PHA}}^\top, \quad (147)$$

$$\bar{v}_{18} = \beta_P(e_{S_{NO2}} - e_{S_{NO3}})^\top + e_{X_{PAO}}^\top - Y_{PAO}^{-1}e_{X_{PHA}}^\top, \quad (148)$$

$$\bar{v}_{19} = \gamma_P(e_{S_{N2}} - e_{S_{NO2}})^\top + e_{X_{PAO}}^\top - Y_{PAO}^{-1}e_{X_{PHA}}^\top, \quad (149)$$

$$\bar{v}_{20} = f_{XI}e_{X_I}^\top + (1 - f_{XI})e_{X_S}^\top - e_{X_{PAO}}^\top, \quad (150)$$

$$\bar{v}_{21} = -e_{X_{PP}}^\top, \quad \bar{v}_{22} = e_{S_A}^\top - e_{X_{PHA}}^\top, \quad (151)$$

$$\bar{v}_{23} = -\frac{3.43 - Y_{AOB}}{Y_{AOB}}e_{S_O}^\top + Y_{AOB}^{-1}e_{S_{NO2}}^\top + e_{X_{AOB}}^\top, \quad (152)$$

$$\bar{v}_{24} = -\frac{1.14 - Y_{NOB}}{Y_{NOB}}e_{S_O}^\top - Y_{NOB}^{-1}e_{S_{NO2}}^\top + Y_{NOB}^{-1}e_{S_{NO3}}^\top + e_{X_{NOB}}^\top, \quad (153)$$

$$\bar{v}_{25} = f_{XI}e_{X_I}^\top + (1 - f_{XI})e_{X_S}^\top - e_{X_{AOB}}^\top, \quad (154)$$

$$\bar{v}_{26} = f_{XI}e_{X_I}^\top + (1 - f_{XI})e_{X_S}^\top - e_{X_{NOB}}^\top, \quad (155)$$

$$\bar{v}_{27} = -e_{S_{PO4}}^\top - \gamma_{OH}e_{X_{MeOH}}^\top + i_{PMeP}^{-1}e_{X_{MeP}}^\top, \quad (156)$$

$$\bar{v}_{28} = e_{S_{PO4}}^\top + \gamma_{OH}e_{X_{MeOH}}^\top - i_{PMeP}^{-1}e_{X_{MeP}}^\top, \quad (157)$$

where

$$\beta_H = \frac{1 - Y_H}{(8/7)Y_H}, \quad \gamma_H = \frac{1 - Y_H}{1.72Y_H}, \quad \beta_S = \frac{Y_{PHA}}{8/7}, \quad \gamma_S = \frac{Y_{PHA}}{1.72}, \quad (158)$$

$$\beta_P = \frac{1 - Y_{PAO}}{(8/7)Y_{PAO}}, \quad \gamma_P = \frac{1 - Y_{PAO}}{1.72Y_{PAO}}. \quad (159)$$

Define

$$\mathcal{N} = \{S_F, S_I, S_{N2}, S_{NO2}, S_{NO3}, X_I, X_S, X_H, X_{PAO}, X_{AOB}, X_{NOB}\}, \quad (160)$$

$$\mathcal{P} = \{S_F, S_I, X_I, X_S, X_H, X_{PAO}, X_{AOB}, X_{NOB}, X_{PP}, X_{MeP}\}. \quad (161)$$

Nitrogen continuity completes every S_{NH4} coefficient:

$$v_{p,S_{NH4}} = - \sum_{j \in \mathcal{N}} n_j \bar{v}_{p,j}, \quad (162)$$

where $n_{S_F} = i_{NSF}$, $n_{S_I} = i_{NSI}$, $n_{S_{N2}} = n_{S_{NO2}} = n_{S_{NO3}} = 1$, $n_{X_I} = i_{NXI}$, $n_{X_S} = i_{NXS}$, and $n_{X_H} = n_{X_{PAO}} = n_{X_{AOB}} = n_{X_{NOB}} = i_{NBM}$. Except for the explicitly specified rows 27–28, phosphorus continuity gives

$$v_{p,S_{PO4}} = - \sum_{j \in \mathcal{P}} p_j \bar{v}_{p,j}, \quad (163)$$

where $p_{S_F} = i_{PSF}$, $p_{S_I} = i_{PSI}$, $p_{X_I} = i_{PXI}$, $p_{X_S} = i_{PXS}$, $p_{X_H} = p_{X_{PAO}} = p_{X_{AOB}} = p_{X_{NOB}} = i_{PBM}$, $p_{X_{PP}} = 1$, and $p_{X_{MeP}} = i_{PMeP}$. All weights not listed are zero. Alkalinity is then

$$v_{p,S_{ALK}} = \frac{v_{p,S_{NH4}}}{14} - \frac{v_{p,S_{NO2}}}{14} - \frac{v_{p,S_{NO3}}}{14} + \frac{v_{p,S_{PO4}}}{31}. \quad (164)$$

For every other coordinate $j \notin \{S_{NH4}, S_{PO4}, S_{ALK}\}$, set $v_{p,j} = \bar{v}_{p,j}$; rows 27–28 retain their displayed S_{PO4} entries. Together with the preliminary rows, these rules define all entries of $v \in \mathbb{R}^{28 \times 20}$.

The deterministic physical conservation rows are

$$b_{SI} = e_{S_I}^\top, \quad (165)$$

$$b_N = e_{S_{NH4}}^\top + e_{S_{NO2}}^\top + e_{S_{NO3}}^\top + e_{S_{N2}}^\top + i_{NSI} e_{S_I}^\top + i_{NSF} e_{S_F}^\top + i_{NXI} e_{X_I}^\top + i_{NXS} e_{X_S}^\top \quad (166)$$

$$+ i_{NBM} (e_{X_H} + e_{X_{PAO}} + e_{X_{AOB}} + e_{X_{NOB}})^\top, \quad (167)$$

$$b_P = e_{S_{PO4}}^\top + i_{PSI} e_{S_I}^\top + i_{PSF} e_{S_F}^\top + i_{PXI} e_{X_I}^\top + i_{PXS} e_{X_S}^\top + e_{X_{PP}}^\top \quad (168)$$

$$+ i_{PBM} (e_{X_H} + e_{X_{PAO}} + e_{X_{AOB}} + e_{X_{NOB}})^\top + i_{PMeP} e_{X_{MeP}}^\top, \quad (169)$$

$$b_{alk} = e_{S_{ALK}}^\top - (1/14) e_{S_{NH4}}^\top + (1/14) e_{S_{NO2}}^\top + (1/14) e_{S_{NO3}}^\top - (1/31) e_{S_{PO4}}^\top, \quad (170)$$

$$b_{metal} = \gamma_{OH} e_{X_{MeP}}^\top + i_{PMeP}^{-1} e_{X_{MeOH}}^\top. \quad (171)$$

With

$$B = \begin{bmatrix} b_{SI} \\ b_N \\ b_P \\ b_{alk} \\ b_{metal} \end{bmatrix} \in \mathbb{R}^{5 \times 20} \quad (172)$$

and b_k denoting row k , define the deterministic row-normalized operator

$$D_A = \text{Diag}(\|b_1\|_2^{-1}, \dots, \|b_5\|_2^{-1}), \quad A = D_A B. \quad (173)$$

Thus $Av^\top = 0$, $Ae_{S_O} = 0$, $\text{rank}(A) = 5$, and every row of A has unit Euclidean norm. No linear combination is taken between the five named inventories.

A.3. Composite maps

In the component order of Equation (40), the rows of the composite map are COD, reported TN, TP, and TSS:

$$I_{\text{comp}} = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0.01 & 0 & 0.02 & 0 & 0.07 & 0.07 & 0 & 0 & 0.07 & 0.07 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0.01 & 0 & 0.02 & 0.02 & 1 & 0 & 0.02 & 0.02 & i_{PMeP} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.75 & 0.75 & 0.90 & 0.90 & 3.23 & 0.60 & 0.90 & 0.90 & 1 \end{bmatrix}. \quad (174)$$

Reported TN excludes S_{N2} because it represents converted dinitrogen rather than a regulated aqueous nitrogen species. The TSS vector is the transpose of the last row:

$$t_X = (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0.75, 0.75, 0.90, 0.90, 3.23, 0.60, 0.90, 0.90, 1)^\top. \quad (175)$$

A.4. Fixed biological and chemical parameters

Table 5: Complete ASM2d-TSN parameter set.

Group	Symbol	Value	Unit
Stoichiometry	Y_H	0.625	g COD g COD^{-1}
	Y_{PAO}	0.625	g COD g COD^{-1}
	Y_{PHA}	0.20	g COD g P^{-1}
	Y_{PO4}	0.40	g P g COD^{-1}
	Y_{AOB}	0.18	g COD g N^{-1}
	Y_{NOB}	0.08	g COD g N^{-1}
	f_{SI}	0	—
	f_{XI}	0.10	—
Composition	i_{NSI}, i_{NSF}	0.01, 0	g N g COD^{-1}
	$i_{NXI}, i_{NXS}, i_{NBM}$	0.02, 0, 0.07	g N g COD^{-1}
	i_{PSI}, i_{PSF}	0, 0	g P g COD^{-1}
	$i_{PXI}, i_{PXS}, i_{PBM}$	0.01, 0, 0.02	g P g COD^{-1}
	i_{PMeP}	1/4.87	g P g TSS^{-1}
	$i_{TSS,XI}, i_{TSS,XS}$	0.75, 0.75	g TSS g COD^{-1}
	$i_{TSS,BM}, i_{TSS,PHA}$	0.90, 0.60	g TSS g COD^{-1}
	$i_{TSS,PP}$	3.23	g TSS g P^{-1}
Hydrolysis	K_H	3.0	d^{-1}
	$\eta_{hyd,NO3}, \eta_{hyd,NO2}, \eta_{hyd,fe}$	0.6, 0.6, 0.4	—
	$K_{O,hyd}$	0.20	$\text{g O}_2 \text{ m}^{-3}$
	$K_{NO3,hyd}, K_{NO2,hyd}, K_{NOx,hyd}$	0.5, 0.5, 0.5	g N m^{-3}
	K_X	0.10	$\text{g } X_S \text{ g } X_H^{-1}$
Heterotrophs	μ_H, q_{fe}, b_H	6.0, 3.0, 0.40	d^{-1}
	$\eta_{H,NO3}, \eta_{H,NO2}$	0.9, 0.9	—
	$K_{O,H}$	0.20	$\text{g O}_2 \text{ m}^{-3}$
	K_F, K_{fe}, K_A	4, 4, 4	g COD m^{-3}
	$K_{NO3,H}, K_{NO2,H}, K_{NOx,H}$	0.5, 0.5, 0.5	g N m^{-3}
	$K_{NH4,H}$	0.05	g N m^{-3}
	$K_{PO4,H}$	0.01	g P m^{-3}
	$K_{ALK,H}$	0.10	$\text{mol HCO}_3^- \text{ m}^{-3}$
PAO	q_{PHA}	5.0	d^{-1}
	q_{PP}	0.60	$\text{g P (g COD d)}^{-1}$
	μ_{PAO}	0.56	d^{-1}
	$\eta_{PAO,NO3}, \eta_{PAO,NO2}$	0.07, 0.90	—
	b_{PAO}, b_{PP}, b_{PHA}	0.20, 0.20, 0.20	d^{-1}
	$K_{O,PAO}$	0.20	$\text{g O}_2 \text{ m}^{-3}$
	$K_{NO3,PAO}, K_{NO2,PAO}, K_{NOx,PAO}$	0.5, 0.5, 0.5	g N m^{-3}
	$K_{NH4,PAO}$	0.05	g N m^{-3}
	$K_{PS}, K_{PO4,PAO}$	0.20, 0.01	g P m^{-3}
	$K_{ALK,PAO}$	0.10	$\text{mol HCO}_3^- \text{ m}^{-3}$
	$K_{PP}, K_{\max}, K_{IPP}$	0.01, 0.34, 0.02	g P g COD^{-1}
	K_{PHA}	0.01	g COD g COD^{-1}
	μ_{AOB}, μ_{NOB}	1.81, 1.52	d^{-1}
	b_{AOB}, b_{NOB}	0.20, 0.17	d^{-1}
	$K_{O,AOB}, K_{O,NOB}$	0.74, 1.75	$\text{g O}_2 \text{ m}^{-3}$
Nitrifiers	$K_{NH4,AOB}, K_{NO2,NOB}$	0.5, 0.5	g N m^{-3}
	$K_{ALK,mit}$	0.5	$\text{mol HCO}_3^- \text{ m}^{-3}$

Group	Symbol	Value	Unit
Chemical	$K_{PO4,nit}$	0.01	g P m^{-3}
	k_{PRE}	1.0	$\text{m}^3 (\text{g TSS d})^{-1}$
	k_{RED}	0.60	d^{-1}
	γ_{OH}	3.45	g TSS g P^{-1}
	$K_{ALK,chem}$	0.50	$\text{mol HCO}_3^- \text{m}^{-3}$

CRedit authorship contribution statement

The sole author, Egberto F. Selerio Jr., contributed to conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft, writing—review & editing, visualization, supervision, project administration, and funding acquisition.

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Data availability

All equations, parameter values, coordinate orders, sampling rules, estimators, acceptance tests, and search rules needed to reproduce the calculation are contained in this article. The generated 20,000-state table, fitted coefficient matrix, assessment table, and complete optimization archives will be deposited as a public research-data record when the numerical study is complete.

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